Universitat Politècnica de Catalunya

Department of Computer Science
Department of Statistics and Operations Research

Research Report

Downstream Utility of Synthetic Data

Structural and Inferential Evaluation in Clustering and Regression

Oriol Farrés Vilar

Supervisor: **Jordi Cortés**

Barcelona, April 27, 2026

Abstract

Synthetic data is increasingly used as a privacy-preserving substitute for original data, but its value depends on whether it preserves the conclusions of downstream analyses. This thesis evaluates that analytical utility in two complementary settings: clustering, where utility depends on structural and geometric fidelity, and regression, where utility depends on preserving coefficients, uncertainty, and confidence intervals.

The first study examines whether synthetic data generated with Classification and Regression Trees (CART) preserves the geometry required by clustering workflows. It compares K-Means and Agglomerative Hierarchical Clustering across controlled simulations varying sample size, cluster count, separation, dimensionality, correlation, and distribution family. The results show that CART remains useful for spherical normal settings but introduces block-like artifacts that degrade centroid-based clustering on skewed Gamma data. Hierarchical Clustering is more robust when synthesis distorts non-convex structure.

The second study evaluates inferential utility for linear and logistic regression. It compares CART-based synthesis with parametric generators using Confidence Interval Overlap, Bias Ratio, and Confidence Interval Width Ratio. Parametric synthesis performs best when its assumptions match the data-generating process, while CART becomes less reliable in high-dimensional, correlated, small-sample, and logistic settings. Overall, the thesis shows that synthetic-data utility is not intrinsic to a generator alone; it depends on the interaction between synthesis method, data structure, and downstream analytical task.

Keywords: Synthetic Data, Clustering, Regression, K-Means, Hierarchical Clustering, CART, Data Utility, Privacy.

Resum

Les dades sintètiques s'utilitzen cada vegada més com a substitut de les dades originals per protegir la privadesa, però el seu valor depèn de si conserven les conclusions de les anàlisis posteriors. Aquesta tesi avalua aquesta utilitat analítica en dos contextos complementaris: el clustering, on la utilitat depèn de la fidelitat estructural i geomètrica, i la regressió, on la utilitat depèn de preservar coeficients, incertesa i intervals de confiança.

El primer estudi analitza si les dades sintètiques generades amb arbres de classificació i regressió (CART) conserven la geometria necessària per a tasques de clustering. Es comparen K-Means i clustering jeràrquic aglomeratiu en simulacions controlades que varien la mida mostral, el nombre de clústers, la separació, la dimensionalitat, la correlació i la família de distribucions. Els resultats mostren que CART és útil en escenaris normals i esfèrics, però introdueix artefactes en blocs que perjudiquen el clustering basat en centroides en dades Gamma esbiaixades. El clustering jeràrquic és més robust quan la síntesi distorsiona estructures no convexes.

El segon estudi avalua la utilitat inferencial en regressió lineal i logística. Es compara la síntesi basada en CART amb generadors paramètrics mitjançant el solapament d'intervals de confiança, el ràtio de biaix i el ràtio d'amplada dels intervals. La síntesi paramètrica obté els millors resultats quan les seves hipòtesis coincideixen amb el procés generador de dades, mentre que CART és menys fiable en escenaris d'alta dimensionalitat, alta correlació, mostres petites i regressió logística. En conjunt, la tesi mostra que la utilitat de les dades sintètiques no depèn només del generador, sinó de la interacció entre el mètode de síntesi, l'estructura de les dades i la tasca analítica posterior.

Paraules clau: Dades Sintètiques, Clustering, Regressió, K-Means, Clustering Jeràrquic, CART, Utilitat de les Dades, Privadesa.

Resumen

Los datos sintéticos se utilizan cada vez más como sustituto de los datos originales para preservar la privacidad, pero su valor depende de si conservan las conclusiones de los análisis posteriores. Esta tesis evalúa esa utilidad analítica en dos contextos complementarios: el clustering, donde la utilidad depende de la fidelidad estructural y geométrica, y la regresión, donde la utilidad depende de preservar coeficientes, incertidumbre e intervalos de confianza.

El primer estudio analiza si los datos sintéticos generados mediante árboles de clasificación y regresión (CART) conservan la geometría necesaria para tareas de clustering. Se comparan K-Means y clustering jerárquico aglomerativo en simulaciones controladas que varían el tamaño muestral, el número de clústeres, la separación, la dimensionalidad, la correlación y la familia de distribuciones. Los resultados muestran que CART es útil en escenarios normales y esféricos, pero introduce artefactos en bloques que perjudican el clustering basado en centroides en datos Gamma sesgados. El clustering jerárquico es más robusto cuando la síntesis distorsiona estructuras no convexas.

El segundo estudio evalúa la utilidad inferencial en regresión lineal y logística. Se compara la síntesis basada en CART con generadores paramétricos usando el solapamiento de intervalos de confianza, el ratio de sesgo y el ratio de anchura de los intervalos. La síntesis paramétrica obtiene mejores resultados cuando sus supuestos coinciden con el proceso generador de datos, mientras que CART es menos fiable en escenarios de alta dimensionalidad, alta correlación, muestras pequeñas y regresión logística. En conjunto, la tesis muestra que la utilidad de los datos sintéticos no depende solo del generador, sino de la interacción entre el método de síntesis, la estructura de los datos y la tarea analítica posterior.

Palabras clave: Datos Sintéticos, Clustering, Regresión, K-Means, Clustering Jerárquico, CART, Utilidad de los Datos, Privacidad.

Acknowledgements

I would like to express my sincere gratitude to my supervisor, Jordi Cortés, for his continuous support, patience, and motivation throughout this journey. It has been an absolute pleasure working with him for more than a year; his guidance, technical expertise, and encouragement have been fundamental to my growth as a researcher and the successful completion of this thesis.

Our collaboration evolved into a remarkably dynamic and ambitious partnership. What began as a focused investigation into clustering utility soon grew into a shared desire to push the boundaries of this research further. This mutual drive led us to expand the scope significantly, venturing into linear and then logistic regression to build a truly comprehensive evaluation. We consistently sought to exceed the initial expectations, resulting in a body of work that far transcends the standard 6-ECTS requirement. It has been a genuine privilege to work alongside someone who not only maintains a relentless pace but often sets it, constantly challenging me to reach higher levels of rigor and dedication. In an environment where finding such a matched tempo is rare, I am extremely proud of what we have achieved together through this constant, mutual push for excellence.

I would also like to thank Xinnuo Chen, whose preliminary exploratory work provided a solid foundation and inspired the systematic evaluation conducted in this study.

Contents

List of Figures

List of Tables

Part I

Introduction

Chapter 1

General Introduction

1.1 Motivation and Context

Data-driven research increasingly depends on access to detailed datasets, yet the same level of detail that makes data useful can also make it sensitive. In areas such as healthcare, finance, public administration, and social science, individual-level records often cannot be shared freely because of legal, ethical, and institutional constraints. Synthetic data has therefore become an important candidate for privacy-preserving analysis: instead of releasing the original records, a generator produces artificial observations intended to preserve the statistical properties needed for research [1]–[3].

The practical value of synthetic data, however, cannot be judged only by whether the generated records look plausible. The central question is whether an analyst using the synthetic data would reach conclusions similar to those obtained from the original data. This is the problem of analytical utility. Prior work distinguishes between broad similarity measures and task-specific utility measures, emphasizing that the right evaluation depends on the downstream analysis being supported [4].

This thesis studies synthetic-data utility from that downstream perspective. It asks whether common synthesis methods preserve enough information for two different families of machine-learning and statistical analysis:

- **clustering**, where the relevant utility is structural and geometric;
- **regression**, where the relevant utility is inferential and concerns coefficients, uncertainty, and confidence intervals.

1.2 Machine-Learning Perspective

Machine learning includes a broad set of methods that learn from data rather than relying on explicitly programmed rules. A common taxonomy distinguishes between supervised learning, unsupervised learning, and reinforcement learning. Figure 1.1 summarizes this classification and positions the two analytical settings studied in this thesis.

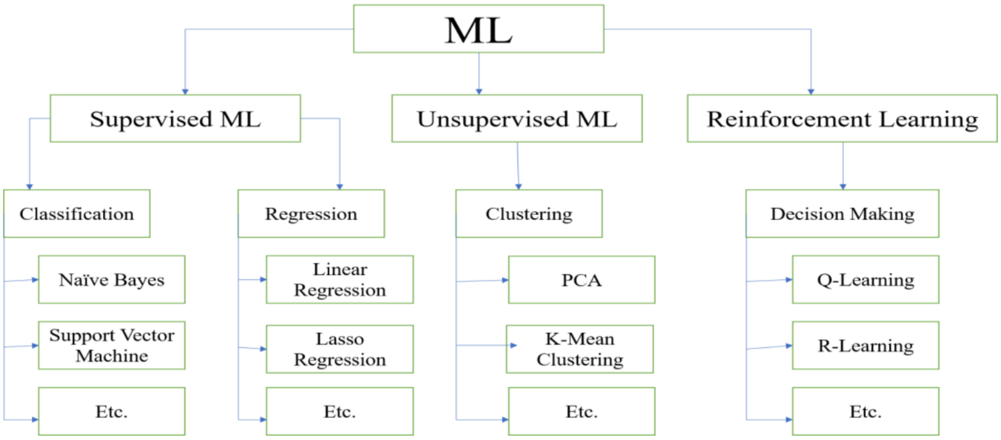

Figure 1.1: Taxonomy of common machine-learning paradigms and representative algorithms.

Source: Compiled by the author

Regression belongs to the supervised or model-based inferential setting, because it studies how an outcome changes as a function of predictors. Clustering belongs to unsupervised learning, because it attempts to discover structure without observing ground-truth labels during training. These two settings place different demands on synthetic data. Clustering requires preservation of distances, densities, separability, and cluster geometry. Regression requires preservation of conditional relationships, point estimates, standard errors, and confidence intervals [5], [6].

1.3 Problem Statement

The motivating assumption behind synthetic data is that a generated dataset can act as a useful substitute for the original data. This thesis treats that assumption as empirical rather than automatic. A synthetic generator may preserve marginal distributions while distorting the multivariate structure required by a downstream method. Conversely, a generator may be adequate for one analysis and unsuitable for another.

The synthesis mechanism is especially important. A major part of this thesis evaluates Classification and Regression Trees (CART), as implemented in the `synthpop` package [3]. CART is flexible because it models variables through data-driven partitions, but those partitions are axis-aligned. As a result, smooth relationships can become block-like or staircase-shaped after synthesis. The key question is not only whether these artifacts exist, but whether they matter for the analysis that follows.

The thesis is therefore organized around a single claim: synthetic-data utility is not an intrinsic property of a generator alone. It emerges from the interaction between the synthesis method, the data-generating structure, and the downstream analytical task.

1.4 Objectives and Research Questions

The general objective of this thesis is to evaluate whether synthetic data preserves downstream analytical utility across two complementary settings: clustering and regression.

The specific objectives are:

1. To evaluate whether CART-generated synthetic data preserves the geometric structure required by clustering algorithms.
2. To compare the robustness of centroid-based and connectivity-based clustering under synthetic distortions.
3. To evaluate whether synthetic data preserves regression coefficients and confidence intervals in linear and logistic models.
4. To compare CART-based synthesis with parametric synthesis when the downstream analysis has a known model structure.
5. To identify conditions under which synthetic data remains useful, becomes unstable, or leads to misleading analytical conclusions.

These objectives are addressed through three thesis-level research questions:

- **RQ1:** Does synthetic data preserve the structural geometry required for unsupervised clustering?
- **RQ2:** Does synthetic data preserve the inferential quantities required for regression analysis?
- **RQ3:** How does the match between generator assumptions, data structure, and downstream analytical method determine synthetic-data utility?

1.5 Structure of the Thesis

The thesis is divided into two empirical studies under this common framing.

Part I studies structural utility for clustering. It evaluates K-Means and Agglomerative Hierarchical Clustering on CART-generated synthetic datasets, using controlled simulations and structural metrics such as cluster-number recovery, Gini coefficient, Mean Centroid Distance, and Mean Variance Difference.

Part II studies inferential utility for regression. It evaluates linear and logistic regression under CART and parametric synthesis, using Confidence Interval Overlap, Bias Ratio, and Confidence Interval Width Ratio to measure whether synthetic data supports the same statistical conclusions as the original data.

The final discussion integrates both studies and presents the overall thesis conclusion: synthetic data should be selected and validated according to the analysis it is expected to support, not treated as a universally interchangeable replacement for original data.

Part II

Clustering Study

Chapter 2

Clustering Utility Study

2.1 Purpose of the Study

Part I evaluates synthetic-data utility in the unsupervised-learning setting. Within the general thesis framework, clustering is used as a test of **structural utility**: whether a synthetic dataset preserves the geometry, separability, density, and dispersion required for an algorithm to recover meaningful groups.

The study focuses on synthetic data generated through the Classification and Regression Trees (CART) method implemented in the `synthpop` package. CART is flexible, but its axis-aligned tree partitions can turn smooth multivariate structure into block-like regions. This potential artifact is directly relevant to clustering because clustering algorithms depend on distances, shapes, and local connectivity.

2.2 Clustering as a Structural Utility Test

Unlike supervised methods, clustering algorithms do not have access to ground-truth labels during training. Their objective is to discover inherent structures by grouping similar observations together based on geometric or statistical properties.

Clustering is fundamental for exploratory data analysis, customer segmentation, and anomaly detection. However, because these algorithms rely heavily on the geometric structure of the data (distances, densities, and distributions), they are highly sensitive to data quality and distribution. Common algorithms include:

- **K-Means:** Partitioning data into k groups by minimizing variance within clusters.
- **Hierarchical Clustering:** Building a tree of clusters (dendrogram) based on distance connectivity.

This part of the thesis extends the work of Chen Xinnuo [7] by systematically comparing how well K-Means and Agglomerative Hierarchical Clustering perform on CART-generated synthetic data.

We aim to answer three specific research questions:

1. **Detection Capability:** Can clustering algorithms correctly identify the optimal number of clusters (k) in synthetic data as well as they do in real data?

2. **Algorithm Robustness:** Which algorithm is more resilient to the distortions introduced by the synthetic generation process? We hypothesise that Hierarchical Clustering may be more robust than K-Means in non-normal distributions.

3. **Quantification of Degradation:** To what extent does the CART generation method degrade the geometric properties (separability, density) of the clusters?

To answer these questions, we have developed a reproducible pipeline that generates 1,800+ datasets varying in sample size (N), cluster separation, noise levels, and probability distributions (Normal vs. Gamma), allowing for a rigorous statistical comparison.

2.3 Structure of Part I

This part is organized as follows:

- **Chapter 3: Methods** details the mathematical formulations of K-Means and Hierarchical Clustering and describes the synthetic data generation via CART. It also defines the structural fidelity metrics and the simulation pipeline used to evaluate algorithm robustness.

- **Chapter 4: Results** presents the empirical analysis, including PCA and t-SNE visualizations alongside cluster detection success rates. It further quantifies geometric degradation using the Gini coefficient, Mean Centroid Distance, and Mean Variance Difference metrics.

- **Chapter 5: Conclusions** synthesizes the clustering findings, highlighting the superior robustness of Hierarchical Clustering for skewed synthetic data and identifying the practical implications for structural utility evaluation.

Chapter 3

Clustering Methods

The methodology adopted in this research is divided into three main stages: the mathematical definition of the clustering algorithms selected for comparison, the generation of synthetic data using the CART method, and the simulation pipeline designed to evaluate the structural degradation of the data.

3.1 Clustering Algorithms

Clustering is the task of grouping a set of objects in such a way that objects in the same group (cluster) are more similar to each other than to those in other groups. This research focuses on two of the most established paradigms in unsupervised learning: centroid-based clustering (K-Means) and connectivity-based clustering (Hierarchical).

3.1.1 K-Means Clustering

K-Means is a partition-based algorithm that aims to divide N observations into k non-overlapping clusters. It operates by minimizing the within-cluster sum of squares (WCSS), also known as inertia [8].

Mathematical Formulation

Given a dataset $X = \{x_1, x_2, ..., x_N\}$ where each $x_i \in \mathbb{R}^d$, the algorithm seeks to partition X into k sets $S = \{S_1, S_2, ..., S_k\}$ to minimize the objective function J:

$$J = \sum_{j=1}^{k} \sum_{x_i \in S_j} ||x_i - \mu_j||^2 \tag{3.1}$$

Where:

- $||x_i - \mu_j||^2$ is the squared Euclidean distance between a data point x_i and the centroid μ_j.
- μ_j is the mean of the points in cluster S_j.

The algorithm proceeds iteratively:

1. **Initialization:** Select k initial centroids randomly.
2. **Assignment:** Assign each observation to the cluster with the nearest centroid.
3. **Update:** Recalculate the centroids as the mean of all observations assigned to each cluster.
4. **Convergence:** Repeat steps 2 and 3 until the centroids do not change significantly or a maximum number of iterations is reached.

Practical Considerations

K-Means is highly efficient with a time complexity of $O(N \cdot k \cdot I \cdot d)$, where I is the number of iterations. However, it has significant limitations relevant to this study:

- It assumes clusters are spherical and of similar size.
- It is sensitive to outliers and the initial placement of centroids.
- The number of clusters k must be specified *a priori*.

3.1.2 Hierarchical Clustering (Agglomerative)

Unlike K-Means, Hierarchical Clustering does not require a pre-specified number of clusters. Instead, it builds a hierarchy of clusters, often represented as a dendrogram. This research utilizes **Agglomerative Clustering** (bottom-up), which treats each observation as its own cluster and successively merges pairs of clusters [9].

Mathematical Formulation (Ward's Linkage)

Agglomerative clustering with Ward's linkage can be framed as a greedy, step-wise optimization problem. Unlike K-Means, which partitions the data, Ward's method builds a hierarchy by performing the optimal local merge at each step.

- **Input:** A set of N vectors $X \subset \mathbb{R}^d$. Initially, each point represents a singleton cluster, resulting in N clusters.
- **Error Sum of Squares (ESS):** For any cluster C, the variance is defined as the sum of squared Euclidean distances from the points in C to its centroid μ_C:

$$\text{ESS}(C) = \sum_{x \in C} ||x - \mu_C||^2 \tag{3.2}$$

- **Objective Function (Step-wise):** At each step, the algorithm finds and merges the pair of clusters (A, B) that results in the minimum increase in the total ESS. The distance (or merging cost) between A and B is exactly this increase:

$$D(A, B) = \text{ESS}_{A \cup B} - (\text{ESS}_A + \text{ESS}_B) \tag{3.3}$$

- **Output:** A dendrogram (tree) representing the sequence of merges, which can be cut at a specific depth to yield k clusters.

Practical Considerations

Hierarchical clustering is computationally expensive, with a time complexity of approximately $O(N^3)$ (or $O(N^2)$ with optimized implementations), making it less suitable for massive datasets. However, it offers distinct advantages:

- It captures hierarchical structures and does not assume strictly spherical clusters.

- It is deterministic (produces the same result every time), unlike K-Means which depends on random initialization.

- It allows for the determination of k *post-hoc* by cutting the dendrogram at a specific height.

3.1.3 Comparative and Complexity Summary

When selecting a clustering algorithm for synthetic data evaluation, understanding the computational constraints is as critical as understanding the geometric assumptions.

Time Complexity: K-Means is highly efficient, with a time complexity of $O(N \cdot k \cdot I \cdot d)$, where N is the number of samples, k is the number of clusters, I is the number of iterations, and d is the dimensionality. Since k, I, and d are usually much smaller than N, it is often considered to have linear time complexity. In contrast, standard Agglomerative Hierarchical Clustering requires computing a distance matrix between all pairs of points, resulting in a time complexity of $O(N^3)$ in the worst case, though optimized implementations (like those using priority queues for Ward's linkage) can reduce this to $O(N^2 \log N)$.

Computational (Space) Complexity: The space complexity of K-Means is $O(N \cdot d + k \cdot d)$ to store the data and the centroids, making its memory requirement grow linearly with the number of samples. Hierarchical Clustering, however, requires $O(N^2)$ space to store the pairwise distance matrix. This quadratic memory requirement makes it computationally prohibitive for very large datasets, presenting a practical tradeoff between its geometric robustness and its resource consumption.

Table 3.1 summarizes the key computational, structural, and operational differences between the two algorithms evaluated in this study.

Table 3.1: Comparison of K-Means and Hierarchical Clustering

Feature	**K-Means**	**Hierarchical (Ward)**
Time Complexity	$O(N \cdot k \cdot I \cdot d)$ (Linear)	$O(N^2 \log N)$ to $O(N^3)$
Space Complexity	$O(N \cdot d + k \cdot d)$ (Linear)	$O(N^2)$ (Quadratic)
Geometry	Assumes spherical, convex clusters	Flexible, based on connectivity
Parameters	Requires k beforehand	k determined by cutting tree
Robustness	Sensitive to noise/outliers	More robust (structure-based)
Deterministic	No (Random initialization)	Yes

Source: Compiled by the author

3.2 Data Generation Framework

To rigorously evaluate algorithm performance, we implemented a reverse-engineering pipeline that generates a "Ground Truth" (Real) dataset and subsequently creates a Synthetic version using the method under investigation.

3.2.1 Reference Data Generation (Real)

The reference datasets were generated using the `mvtnorm` package in R [10], [11] to simulate controlled multivariate clusters. We varied four key parameters to create a total of 180 unique data configurations:

- **Sample Size (N):** Ranging from 100 to 1,000 observations to test scalability.
- **Number of Clusters (k):** Set to 2, 3, or 4 to simulate varying levels of complexity.
- **Cluster Separation:** A separation index controlling the overlap between clusters. Higher separation implies distinct, easy-to-detect clusters; lower separation implies significant overlap.
- **Probability Distribution:**
  - **Normal:** Standard Gaussian clusters (spherical).
  - **Gamma:** Skewed distributions (non-spherical) to test the robustness of the algorithms against non-normal geometries [12].

3.2.2 Synthetic Data Generation (CART Method)

For each real dataset, a corresponding synthetic dataset was generated using the **Classification and Regression Trees (CART)** method, implemented in the `synthpop` R package [3].

The Sequential Regression Approach

The CART method synthesizes data variable by variable. Let $X = (X_1, X_2, ..., X_p)$ be the vector of variables in the real dataset. The joint distribution is approximated as a product of conditional distributions:

$$f(X_1, ..., X_p) \approx f(X_1) \cdot f(X_2|X_1) \cdot f(X_3|X_1, X_2) \cdot ... \cdot f(X_p|X_1, ..., X_{p-1}) \tag{3.4}$$

The synthesis proceeds sequentially:

1. **Step 1:** Generate X_1 by resampling from the original univariate distribution.
2. **Step 2:** Train a Decision Tree (CART) to predict X_2 using X_1 as the predictor.
3. **Step 3:** Generate synthetic values for X_2 by passing the synthetic X_1 through the tree and sampling from the appropriate leaf node.
4. **Step 4:** Repeat for all subsequent variables, using all previously generated variables as predictors.

Structural Artifacts

While CART is non-parametric and can capture non-linear relationships, it introduces specific geometric artifacts. Because decision trees split data using orthogonal hyperplanes (rules like $x > 5$), they tend to approximate smooth diagonal curves with "staircase" or "blocky" boundaries. A key objective of this study is to determine if this "blocky" structure causes K-Means or Hierarchical Clustering to fail.

3.3 Evaluation Metrics

Standard clustering accuracy metrics (like purity) are insufficient when the cluster labels are unknown or permuted. We introduced a set of structural fidelity metrics to quantify the degradation between Real and Synthetic data.

3.3.1 Cluster Alignment (Hungarian Algorithm)

Clustering algorithms are unsupervised; they assign arbitrary labels (e.g., Cluster A found by K-Means might correspond to Cluster B in the ground truth). To calculate accuracy, we first solve the **Linear Sum Assignment Problem** using the **Hungarian Algorithm** [13].

We construct a cost matrix C where C_{ij} represents the number of mismatches if Predicted Cluster i is assigned to True Cluster j. The algorithm finds the optimal permutation of labels that maximizes the overlap between prediction and truth.

3.3.2 Success Rate (Detection)

The primary metric is binary: **Did the algorithm find the correct number of clusters?** We utilize the *Silhouette Score and the Elbow Method* (for K-Means) [14] or the *Dendrogram Cut* (for Hierarchical) to estimate $\hat{k}$.

$$\text{Success} = \begin{cases} 1 & \text{if } \hat{k}_{\text{predicted}} = k_{\text{true}} \\ 0 & \text{otherwise} \end{cases} \tag{3.5}$$

3.3.3 Structural Fidelity Metrics

To measure the geometric distortion introduced by synthesis, we calculate the following comparative metrics between the Real and Synthetic clusters:

1. Gini Coefficient (Cluster Balance)

The Gini coefficient measures the inequality of cluster sizes. If the synthetic data generation collapses a small cluster into a large one, the Gini coefficient will increase.

$$G = \frac{\sum_{i=1}^{k} \sum_{j=1}^{k} |n_i - n_j|}{2k \sum_{i=1}^{k} n_i} \tag{3.6}$$

Where n_i is the number of points in cluster i. A value of 0 indicates perfectly equal cluster sizes; a value near 1 indicates high imbalance.

2. Mean Centroid Displacement (MCD)

We measure how far the centers of the clusters have shifted during synthesis. Let $\mu_j^{(R)}$ be the centroid of cluster j in Real data and $\mu_j^{(S)}$ in Synthetic data:

$$\text{MCD} = \frac{1}{k} \sum_{j=1}^{k} ||\mu_j^{(R)} - \mu_j^{(S)}||_2 \tag{3.7}$$

High displacement indicates that the synthetic data has drifted from the original spatial location.

3. Mean Variance Difference (MVD)

We measure if the synthetic clusters are more dispersed (noisy) or more compact than the real ones. We compare the average within-cluster variance (σ^2):

$$\text{MVD} = \frac{1}{k} \sum_{j=1}^{k} \left| \text{Var}(C_j^{(R)}) - \text{Var}(C_j^{(S)}) \right| \tag{3.8}$$

Chapter 4

Clustering Results

4.1 Visualization and Dimensionality Reduction

4.1.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) projects the data linearly, preserving the global geometric structure and highlighting the orthogonal "blocky" artifacts introduced by the decision-tree-based synthesis [15].

Normal Distribution

Figure 4.1: **PCA Projection: Normal Distribution** $(N = 1000, k = 2)$**.** Comparison of Real (Top) vs. Synthetic (Bottom).

Source: Compiled by the author

In the Normal distribution case, the spherical nature of the clusters is well-preserved. *Observation:* The CART synthesis method preserves the spherical geometry of the Normal distribution (Bottom-Left). Consequently, both K-Means and Hierarchical Clustering successfully recover the true cluster labels, demonstrating high utility for normally distributed synthetic data.

Gamma Distribution

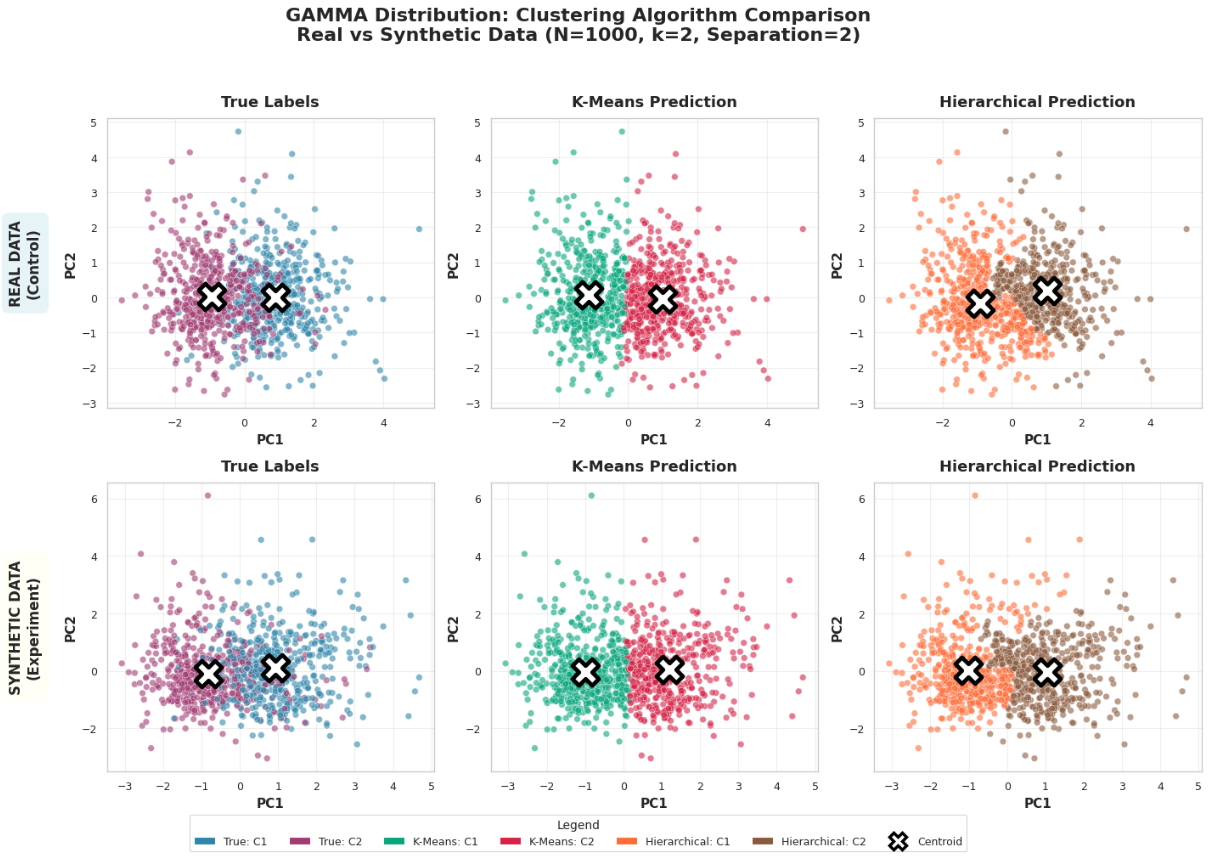

Figure 4.2: **PCA Projection: Gamma Distribution ($N = 1000, k = 2$).** Comparison of Real (Top) vs. Synthetic (Bottom).

Source: Compiled by the author

In the Gamma distribution case, the linear projection reveals the geometric degradation. *Observation:* The synthetic data (Bottom-Left) exhibits "hardened" orthogonal boundaries compared to the smooth tails of the real data. K-Means fails to adapt to this distorted geometry, imposing a linear decision boundary that misclassifies the cluster tails, whereas Hierarchical Clustering proves more robust.

4.1.2 t-SNE Analysis

t-SNE provides a non-linear projection that emphasizes the separation between manifolds [16]. This visualization is particularly effective at highlighting the "bad cuts" made by centroid-based algorithms on skewed data.

Normal Distribution

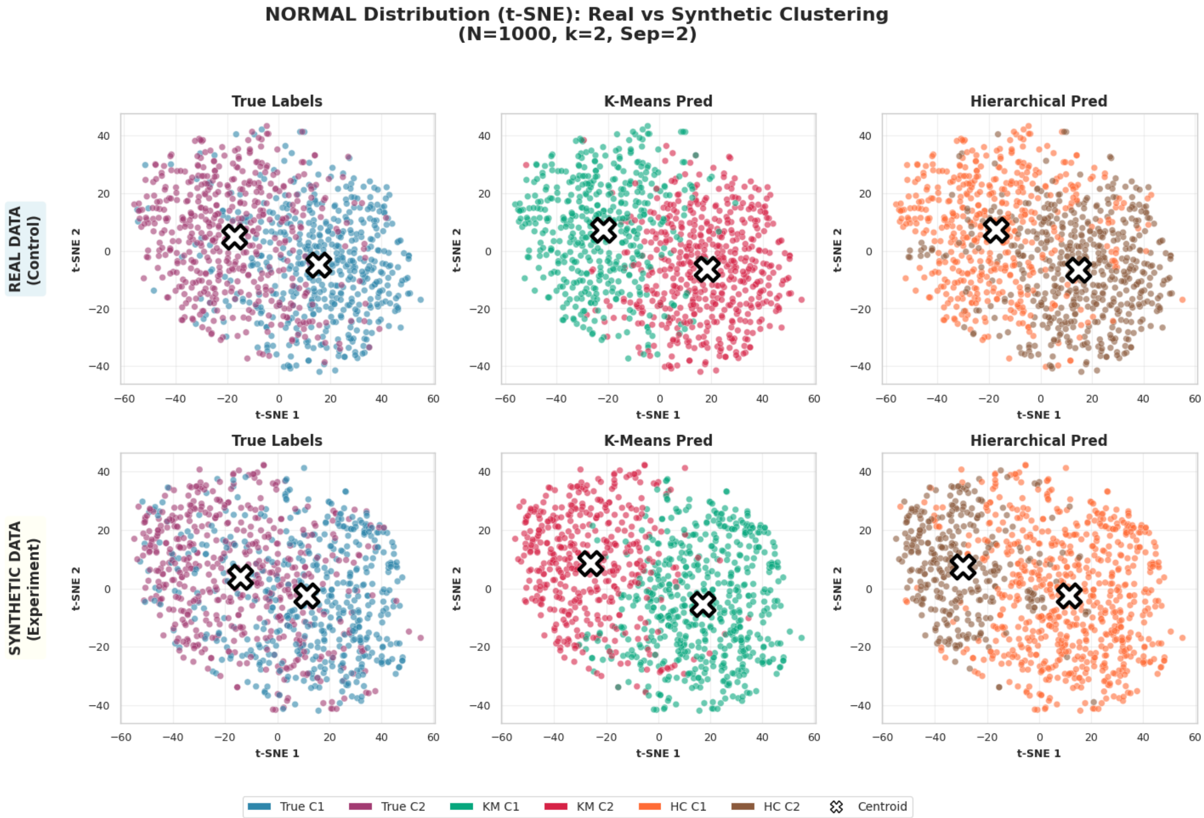

Figure 4.3: **t-SNE Projection: Normal Distribution.** Comparison of Real (Top) vs. Synthetic (Bottom).

Source: Compiled by the author

Even in the non-linear projection, the Normal distribution remains cohesive and separable. *Observation:* The synthetic data (Bottom Row) maintains distinct, separable clusters similar to the control group. Both K-Means and Hierarchical clustering produce nearly identical partitions, confirming that the synthesis process does not introduce topological distortions that affect separability in spherical distributions.

Gamma Distribution

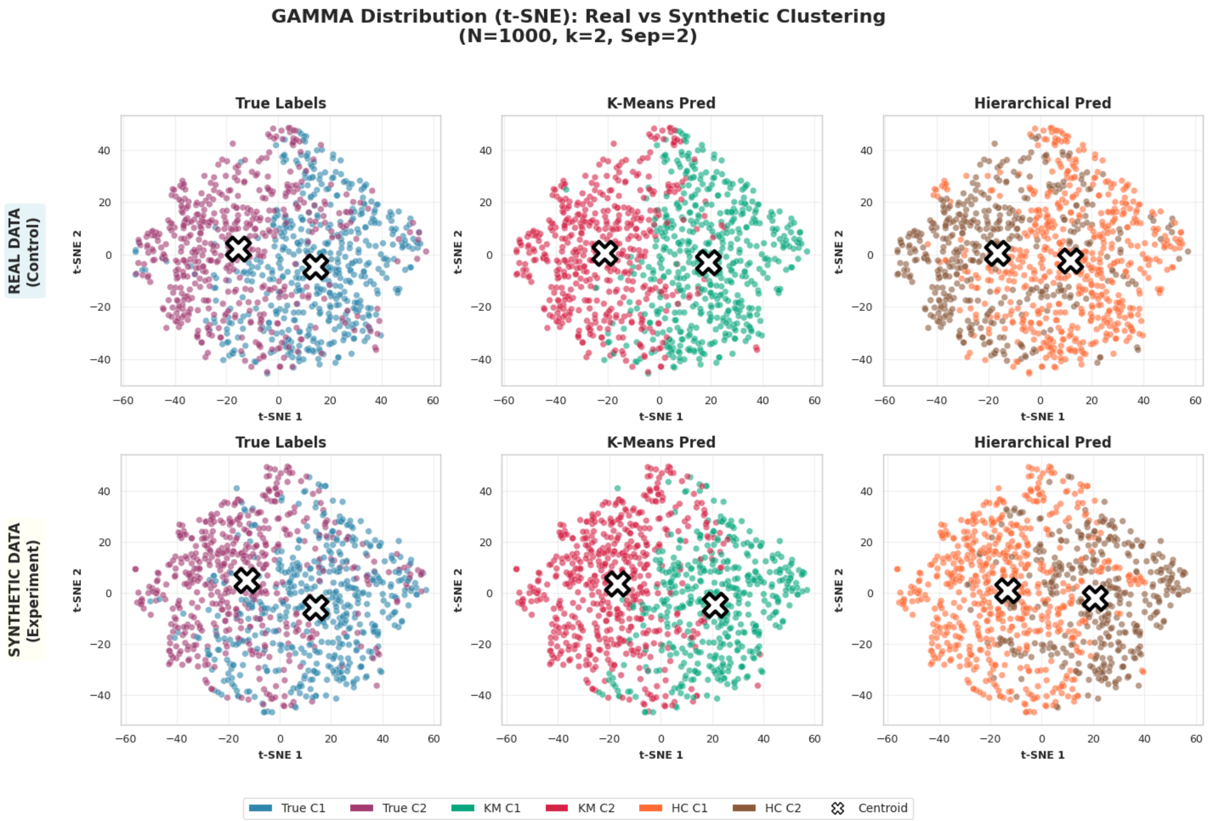

Figure 4.4: **t-SNE Projection: Gamma Distribution.** Comparison of Real (Top) vs. Synthetic (Bottom).

Source: Compiled by the author

The t-SNE projection reveals the catastrophic failure mode of K-Means on skewed synthetic data. *Critical Finding:* On synthetic data (Bottom Row), the True Labels show two connected density masses. K-Means (Center) arbitrarily slices these masses with a rigid boundary, failing to capture the organic shape. In contrast, Hierarchical Clustering (Right) successfully identifies the connected structures, offering a significantly closer approximation to the Ground Truth.

4.2 Cluster Number Matching

This section analyzes the "Success Rate" metric—defined as the proportion of synthetic datasets where the algorithm correctly estimated the true number of clusters (k). We compare the performance of K-Means and Hierarchical Clustering across varying cluster separations and cluster counts.

For a given configuration with M total simulations, the **Success Rate (SR)** is calculated as the proportion of correct detections.

Normal Distribution

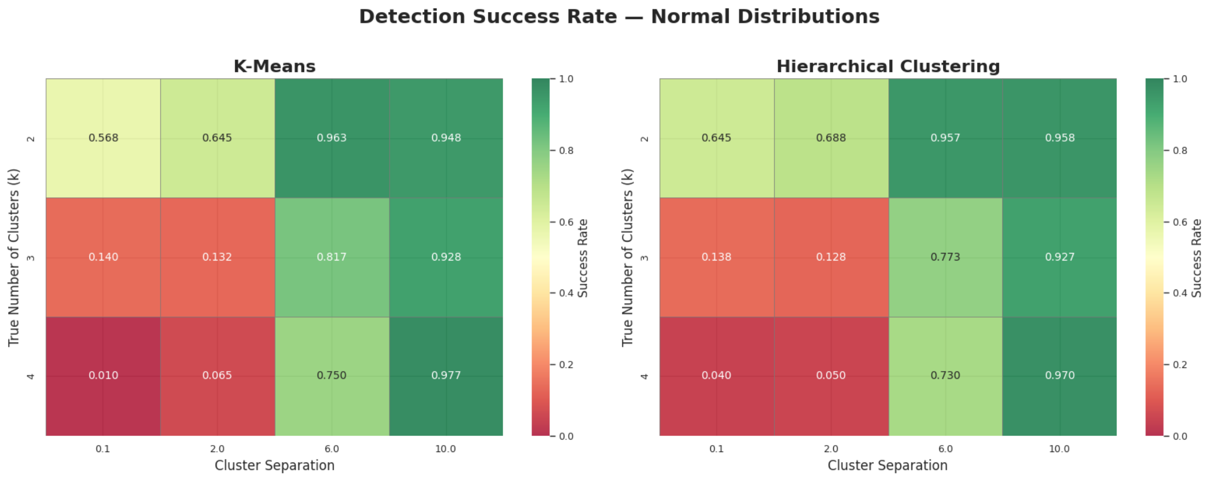

Figure 4.5: **Success Rate on Normal Distribution.** Left: K-Means. Right: Hierarchical Clustering. Green indicates high success; Red indicates failure.

Source: Compiled by the author

Observations:

- **High Separation Stability:** For well-separated clusters, both algorithms achieve near-perfect performance. The synthetic generation artifacts do not hinder detection when the clusters are naturally distinct.

- **Low Separation Degradation:** A sharp performance drop is observed when separation is very low. This suggests that for spherical, overlapping clusters, the "blocky" noise from synthetic generation affects both algorithms equally.

- **Algorithm Parity:** In this distribution, the color patterns between K-Means (Left) and Hierarchical Clustering (Right) are largely symmetrical, indicating no significant advantage for either method on spherical data.

Gamma Distribution

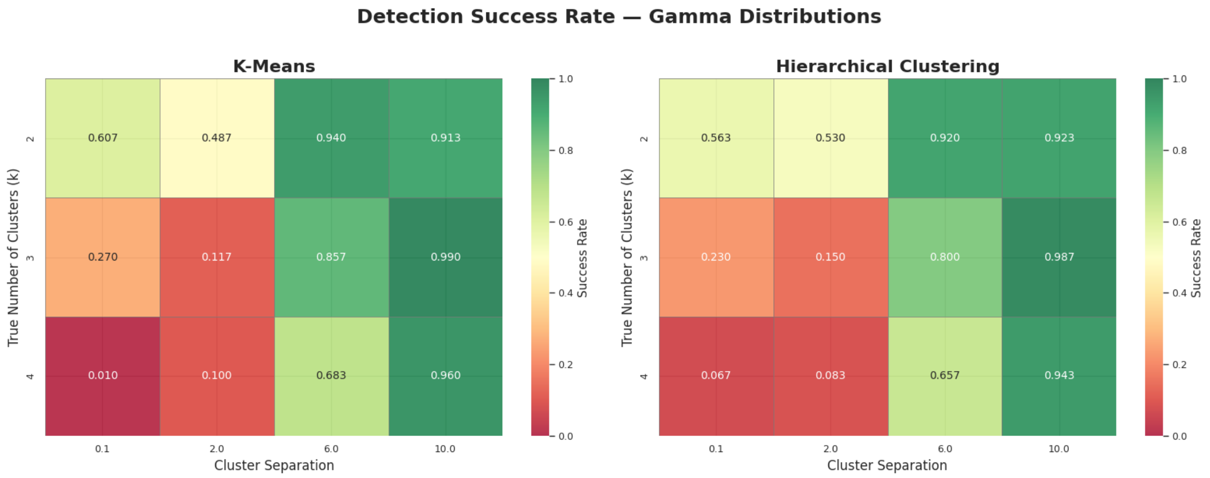

Figure 4.6: **Success Rate on Gamma Distribution.** Left: K-Means. Right: Hierarchical Clustering. Note the prevalence of red/orange zones in the K-Means heatmap.

Source: Compiled by the author

Observations:

- **K-Means Collapse:** The K-Means heatmap exhibits a dominant "Red/Orange" zone extending into moderate separation levels. Detection is not guaranteed even at higher separations.
- **Hierarchical Robustness:** In contrast, the Hierarchical Clustering heatmap retains significantly more "Green" zones, maintaining near-perfect detection while K-Means still shows variability.
- **Performance Delta:** Comparing the two matrices visually reveals a structural divergence. For skewed data, Hierarchical Clustering consistently provides a higher probability of recovering the true k, validating its superior handling of the non-linear artifacts introduced by the synthesis process.

4.3 Gini Coefficient

To assess structural fidelity, we analyzed the Gini coefficient, which measures the inequality of cluster sizes.

Normal Distribution

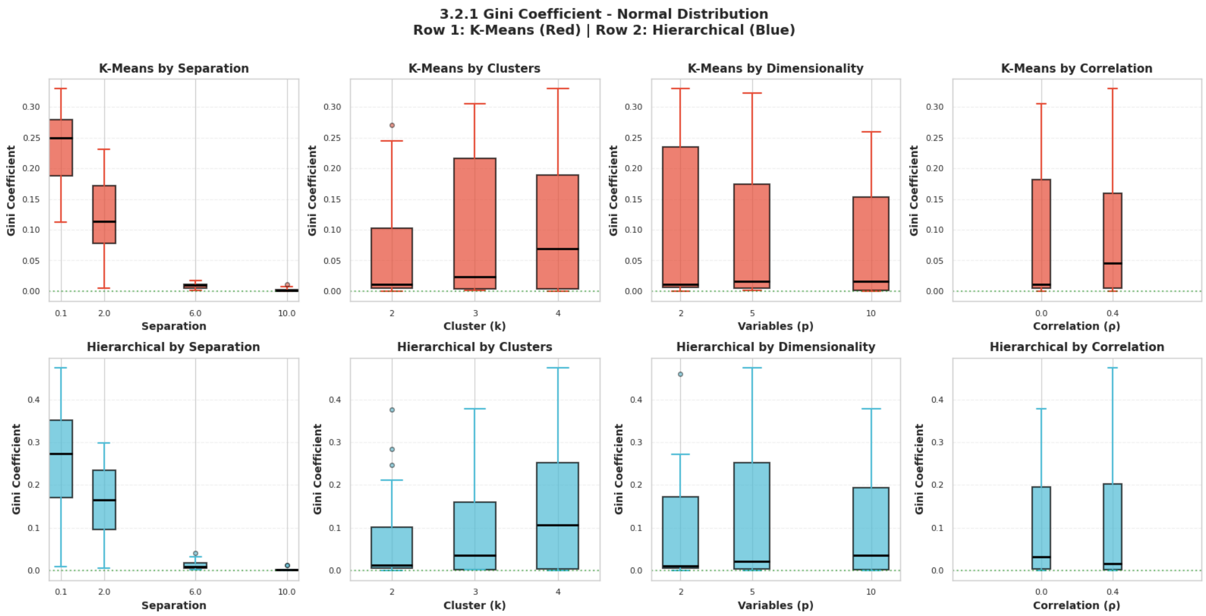

Figure 4.7: **Absolute Gini Coefficients (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric approaches zero as separation increases, indicating balanced detection.

Source: Compiled by the author

Sensitivity to Separation: Both algorithms exhibit a strong dependency on cluster distinctness. At low separation, the synthetic noise dominates, leading to arbitrary partitions and high imbalance. As separation increases, both algorithms converge to near-perfect balance, confirming that the synthetic generation process does not inherently disrupt cluster balance for well-separated spherical data.

Dimensionality Stability: While K-Means maintains a tight distribution near zero for high dimensions, Hierarchical Clustering exhibits a wider interquartile range and a higher median Gini, suggesting less stability in maintaining balance as feature space grows.

Gamma Distribution

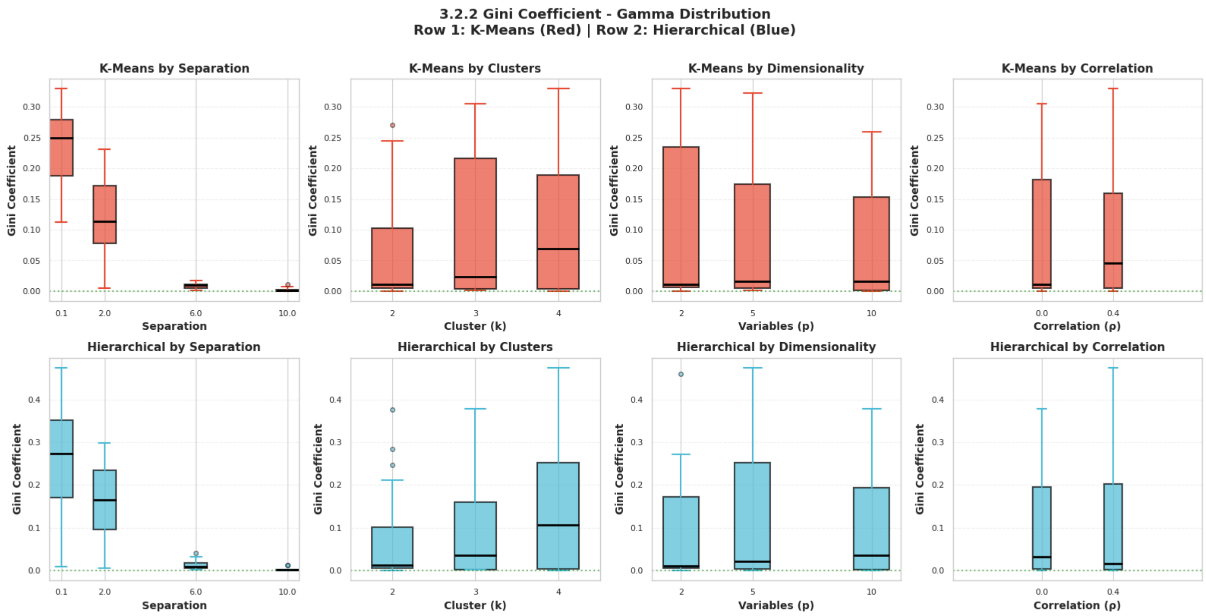

Figure 4.8: **Absolute Gini Coefficients (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). Note the high variance in Hierarchical Clustering compared to the rigid stability of K-Means.

Source: Compiled by the author

K-Means Rigidity: A striking observation is the consistently low Gini coefficients produced by K-Means. Despite the data being skewed and non-spherical, K-Means minimizes within-cluster variance by imposing a Voronoi tessellation that slices the data into roughly equal volumes. This effectively "forces" the skewed data into a balanced structure, suppressing the natural geometric properties of the Gamma distribution.

Hierarchical Flexibility (and Instability): In contrast, Hierarchical Clustering exhibits significantly higher Gini coefficients and wider interquartile ranges. This indicates that Hierarchical Clustering is less constrained by the assumption of equal cluster sizes, allowing it to form imbalanced groups that may better reflect the dense "heads" and diffuse "tails" of the Gamma distribution, though at the cost of higher variance.

Comparative Analysis: While K-Means remains structurally insensitive to the changing geometry (maintaining a flat, low-Gini profile), Hierarchical Clustering reacts dynamically. This confirms that for non-normal synthetic data, **K-Means introduces a "balancing bias"**, while Hierarchical Clustering retains the underlying (imbalanced) structural signal.

4.4 Mean Centroid Distance

We computed the Mean Centroid Distance (MCD) to quantify the spatial shift between the cluster centers in the real dataset versus the synthetic dataset.

Normal Distribution

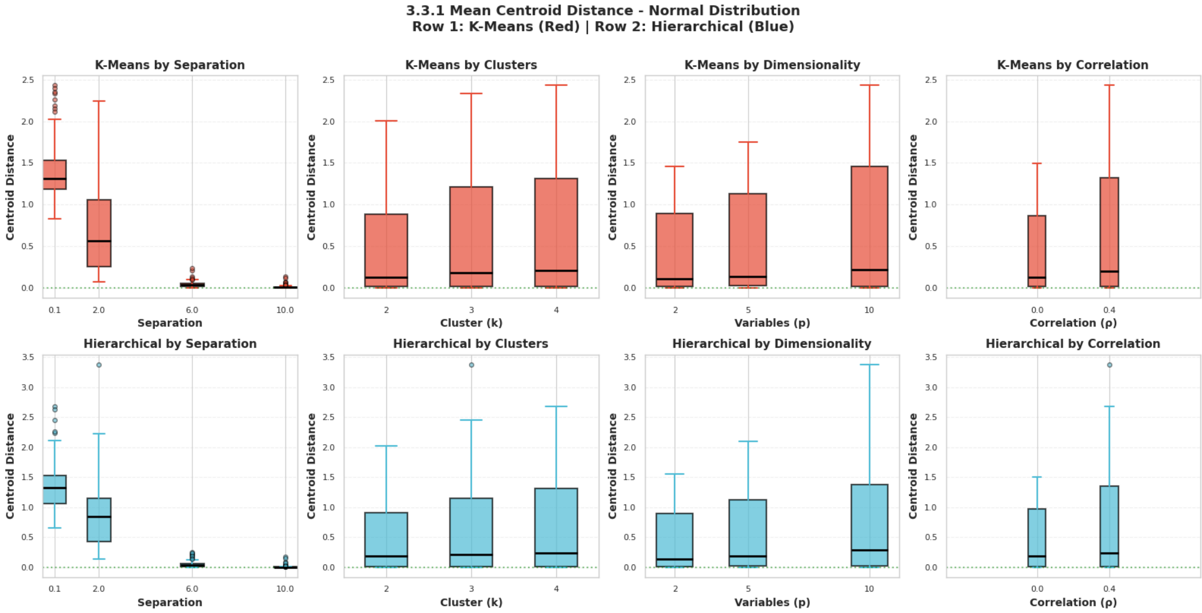

Figure 4.9: **Mean Centroid Distance (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric quantifies the Euclidean shift of cluster centers after synthesis.

Source: Compiled by the author

The results show a dominant inverse relationship between cluster separation and centroid stability. At the lowest separation level, both algorithms show high displacement, indicating significant centroid drift when clusters overlap. As separation increases, the distances for both algorithms effectively collapse to zero. In terms of data complexity, the variance of the centroid distance increases with both the number of clusters (k) and dimensionality (p).

Gamma Distribution

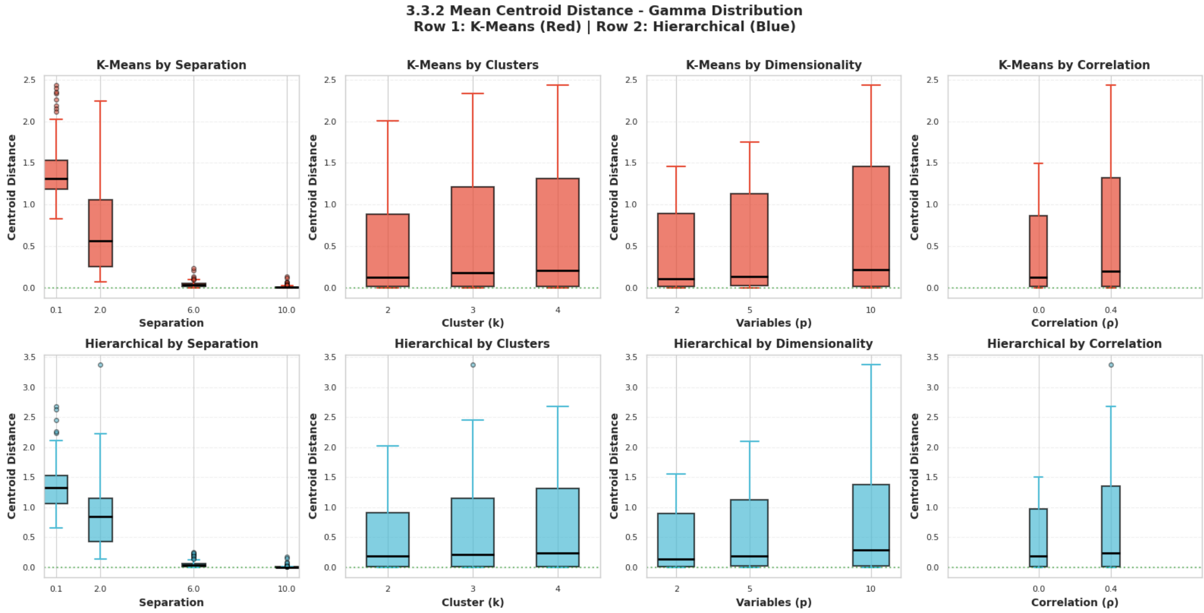

Figure 4.10: **Mean Centroid Distance (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue).

Source: Compiled by the author

The overall trends closely mirror those observed in the Normal distribution, suggesting that centroid drift is primarily driven by the "blocky" artifacts of the CART synthesis method rather than the specific probability distribution. However, the impact of complexity is more pronounced here. Higher dimensionality and a higher number of clusters coincide with increased centroid displacement.

4.5 Mean Variance Difference

We calculated the Mean Variance Difference (MVD) to measure the discrepancy in within-cluster dispersion between the real and synthetic datasets.

Normal Distribution

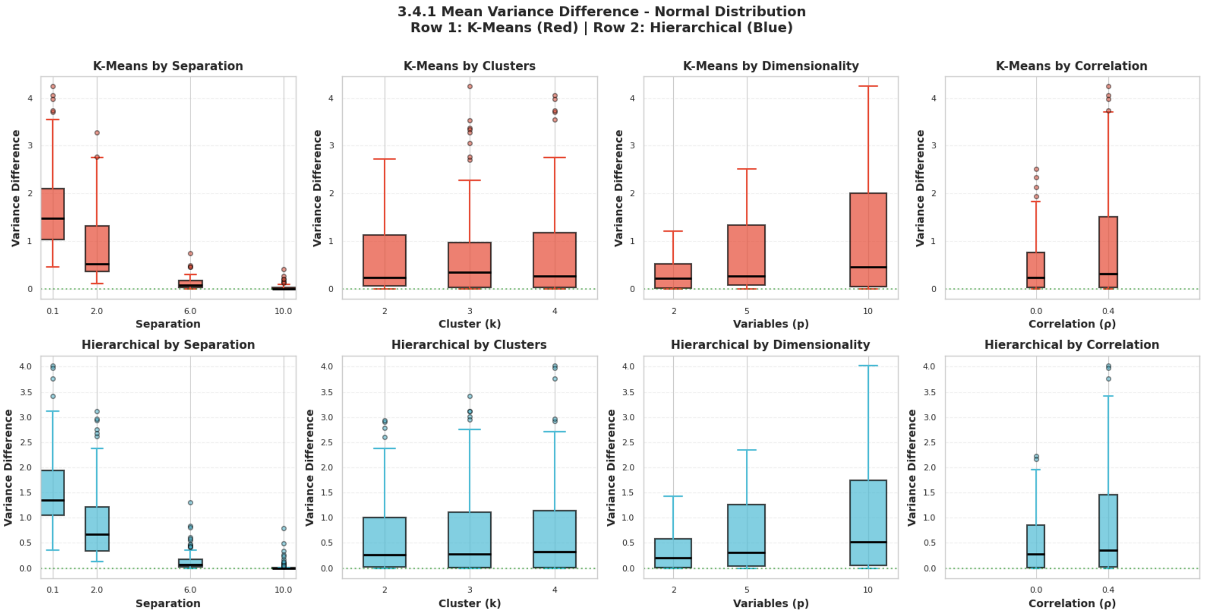

Figure 4.11: **Mean Variance Difference (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric increases with dimensionality and feature correlation.

Source: Compiled by the author

The results indicate that cluster separation is the primary factor reducing variance distortion; at high separation, the difference approaches zero for both algorithms. However, structural complexity negatively impacts fidelity. Increasing the number of variables or introducing correlation significantly increases the variance difference. This suggests that the synthetic generation process struggles to replicate the exact dispersion of multivariate normal clusters in higher-dimensional or correlated spaces.

Gamma Distribution

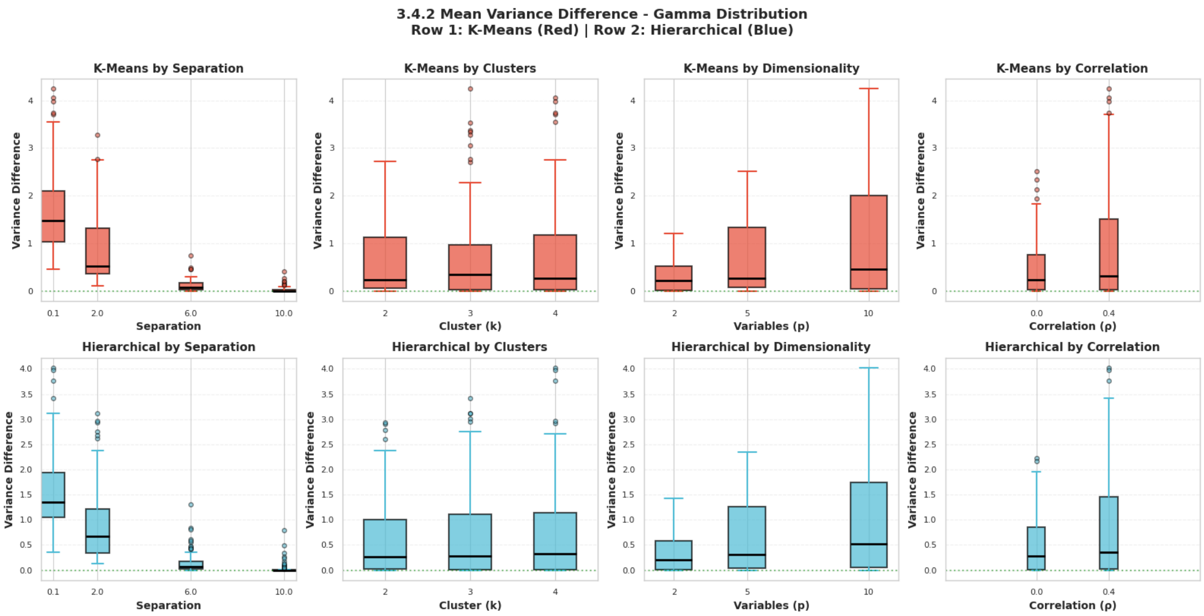

Figure 4.12: **Mean Variance Difference (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). Similar to the Normal case, variance distortion peaks in high-dimensional and correlated settings.

Source: Compiled by the author

The variance discrepancy patterns closely track those observed in the Normal distribution, indicating that dispersion artifacts are driven more by the synthetic generation method than the underlying probability distribution. Furthermore, the “by Dimensionality” and “by Feature Correlation” panels confirm that maintaining accurate cluster variance is increasingly difficult as complexity grows, leading to elevated difference metrics for both algorithms.

Chapter 5

Clustering Conclusions

This research set out to evaluate the "Privacy vs. Utility" trade-off in the context of unsupervised learning, specifically questioning whether synthetic data generated via the Classification and Regression Trees (CART) method retains sufficient geometric fidelity for accurate clustering. By analyzing over 1,800 datasets across varying distributions, noise levels, and dimensionalities, we have quantified the structural degradation introduced by synthesis and its differential impact on K-Means and Hierarchical Clustering.

5.1 Key Findings

The empirical results support the primary hypothesis that the choice of clustering algorithm significantly dictates the utility of synthetic data. Our analysis revealed that the CART synthesis method introduces specific "blocky" orthogonal artifacts due to the nature of decision tree splits. While these artifacts are negligible in spherical Normal distributions, they cause severe topological distortion in skewed Gamma distributions, effectively hardening the smooth tails of clusters into rigid stair-step boundaries.

Consequently, K-Means proved highly sensitive to these artifacts. In skewed environments, the algorithm failed to identify the correct number of clusters even at moderate separation levels, incorrectly imposing linear decision boundaries on the distorted, non-linear synthetic clusters. In contrast, Hierarchical Clustering (Ward's linkage) demonstrated superior robustness, consistently outperforming K-Means in these skewed environments. Its connectivity-based approach allowed it to navigate the non-convex shapes created by the synthesis process, maintaining high detection rates where centroid-based methods collapsed.

Furthermore, our structural fidelity metrics revealed a distinct "balancing bias" in K-Means. The Gini coefficient analysis showed that K-Means artificially forces skewed synthetic data into equal-sized partitions, effectively suppressing the natural imbalance of the data structure. Conversely, Hierarchical Clustering preserved the structural blockiness and skew introduced by the synthesis method. However, both algorithms suffered from centroid drift and variance distortion in high-dimensional or overlapping environments, confirming that the synthesis process itself struggles to replicate complex multivariate correlations regardless of the downstream algorithm used.

5.2 Comparison with previous Work

This study extends the foundational work of Chen Xinnuo [7], which established the baseline utility of synthetic data. While previous studies largely focused on supervised learning tasks, such as classification accuracy, or global statistical properties, this research specifically addressed the unsupervised domain. Unlike general utility metrics that average performance over a dataset, our introduction of structural metrics—specifically the Mean Centroid Distance and Gini Coefficient—allowed for a granular diagnosis of why algorithms fail. We demonstrated that utility is not a binary property of the data, but an interaction between the generation method and the analysis method. Specifically, we identified that the orthogonal constraints of CART are fundamentally incompatible with the spherical convexity assumptions of K-Means when the underlying distribution is non-normal.

5.3 Limitations

The scope of this research implies certain limitations that must be considered when interpreting the results. First, we exclusively evaluated the CART method implemented in the `synthpop` package. Other synthesis techniques, such as parametric methods or deep learning approaches, may produce different geometric artifacts that could interact differently with clustering algorithms. Second, the comparison was strictly limited to K-Means and Agglomerative Hierarchical Clustering. Density-based algorithms like DBSCAN or model-based clustering methods were not tested, though they may offer distinct advantages in handling synthetic noise. Finally, the use of pure Normal and Gamma distributions provided a controlled experimental setting but may not fully capture the chaotic noise, outliers, and mixed-type variables often found in real-world administrative data.

5.4 Generalization

Despite the specific algorithms used, the findings generalize to the broader class of centroid-based versus connectivity-based learning. We posit that any algorithm relying on convex, distance-minimizing objective functions—such as K-Medoids or Linear Discriminant Analysis—will suffer similar degradation when applied to CART-generated synthetic data of skewed distributions. Conversely, methods that prioritize local neighborhood structures, such as Nearest Neighbors or Single-Linkage Clustering, are likely to remain robust to the blocky discontinuities introduced by decision-tree synthesis. This distinction is crucial for data scientists selecting tools for privacy-preserving analytics, as it suggests that the “algorithm fit” is just as critical as the data quality itself.

5.5 Future Research Directions

To further resolve the friction between data privacy and analytical utility, future research should focus on alternative generation methods. Evaluating whether Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs) can generate smooth, non-orthogonal decision boundaries would be a logical next step to see if they preserve the convexity required for K-Means. Additionally, there is significant potential in developing metric-optimized synthesis loss functions that explicitly penalize the distortion of centroid distances or cluster variance during the training of the generator. Finally, creating a standardized “Unsupervised Utility Score” based on the structural metrics defined in this study could help automatically flag synthetic datasets that are unsuitable for clustering tasks before they are released to researchers.

5.6 Conclusions

The promise of synthetic data is that it offers “privacy by design” without compromising analytical truth. This research clarifies the boundaries of that promise. We conclude that synthetic data generated via CART is not utility-neutral; it imparts a structural bias that favors connectivity-based algorithms over centroid-based ones.

For organizations leveraging synthetic data, we recommend a clear operational protocol: if the underlying data distribution is unknown or suspected to be skewed, Hierarchical Clustering must be preferred over K-Means to avoid drawing erroneous conclusions from privacy-preserved data. K-Means should be reserved strictly for scenarios where normality can be verified or where the synthesis method is known to preserve spherical convexity. By understanding these geometric interactions, researchers can effectively unlock the value of sensitive data while maintaining rigorous privacy standards.

Part III

Regression Study

Chapter 6

Regression Utility Study

6.1 Regression as an Inferential Utility Test

Part II evaluates synthetic-data utility in the regression setting. Within the general thesis framework, regression is used as a test of **inferential utility**: whether synthetic data preserves the statistical conclusions that would be obtained from the original data.

Regression analysis remains the most widely used statistical tool for inferential research. Researchers rely on linear and logistic models to quantify relationships, test hypotheses, and inform policy. If synthetic data is to be used as a proxy for real data in academic and professional publications, it must preserve not only the marginal distributions but also the conditional relationships (coefficients) and their associated uncertainty (standard errors and confidence intervals) [5], [6].

Most synthetic data evaluations focus on utility through predictive accuracy or distributional similarity. However, for inference, the primary concern is whether the same scientific conclusion would be reached on synthetic data as on original data [4]. This requires specialized metrics such as Confidence Interval Overlap (CIO) and a systematic understanding of how data complexity, sample size, dimensionality, and correlation affect inferential fidelity.

6.2 Objectives and Research Questions

The primary objective of this part of the study is to evaluate the inferential utility of synthetic data generators across a wide range of simulated regression scenarios, determining which methods best preserve the coefficients and confidence intervals of the original data.

The specific objectives are:

1. To compare the performance of CART-based synthesis against parametric generators (Normal, Logistic) in linear and logistic regression models.

2. To quantify the impact of sample size (N), number of predictors (p), and correlation structure (ρ) on the quality of synthetic inference.

3. To identify the specific conditions under which synthetic data leads to biased or overly optimistic/pessimistic conclusions.

The regression study is guided by the following research questions:

- **RQ1:** How reliably can synthetic data reproduce the confidence intervals of a linear regression model compared to a logistic one?

- **RQ2:** Does increasing the dimensionality (p) or complexity of the correlation structure (ρ) significantly degrade the CIO?

- **RQ3:** Is CART-based synthesis universally superior to parametric methods, or are there specific regimes where it fails?

6.3 Scope and Contributions

This study covers univariate and multivariate simulations for linear and logistic regression. It evaluates synthesis methods provided by the `synthpop` package, specifically focusing on the CART (Classification and Regression Trees), Normal, and Logistic regression synthesis arms.

The study focuses exclusively on **sequential synthetic data generation (1 to 1)**, meaning variables are synthesized one by one conditioned on previously generated ones. It does not cover joint/simultaneous generation methods, deep learning approaches (GANs, VAEs), or explicit privacy-risk measurements (e.g., membership inference attacks).

The study provides a systematic benchmarking of common synthesis methods across 1,800+ scenarios. It introduces a structured evaluation framework based on CIO, Bias Ratio, and Width Ratio, offering practical guidance for researchers on which generator to choose depending on their data's characteristics.

6.4 Structure of Part II

This part is organized as follows: Chapter 7 describes the experimental design and metrics; Chapter 8 presents the results for linear models; Chapter 9 details the logistic regression findings; and Chapter 10 synthesizes the regression conclusions.

Chapter 7

Regression Methods

7.1 Regression Framework

7.1.1 Linear Regression

The linear regression model is the foundation for analyzing the relationship between a continuous outcome Y and a set of predictors $\mathbf{X} = \{X_1, X_2, \ldots, X_p\}$ [17]. The model is defined as:

$$Y = \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{7.1}$$

where β_j are the regression coefficients and ϵ represents the error term.

7.1.2 Logistic Regression

For binary outcomes $Y \in \{0, 1\}$, we employ the logistic regression model [18]:

$$\text{logit}(P(Y = 1|\mathbf{X})) = \ln\left(\frac{P(Y = 1|\mathbf{X})}{1 - P(Y = 1|\mathbf{X})}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j \tag{7.2}$$

7.1.3 Inferential Quantities of Interest

Our evaluation focuses on the preservation of the estimated coefficients $\hat{\beta}_j$ and their associated 95% confidence intervals (CIs). If synthetic data is to be used as a proxy for real data, the synthetic coefficients $\hat{\beta}_j^{(S)}$ should be unbiased estimates of the original coefficients $\hat{\beta}_j^{(R)}$, and their CIs should overlap substantially.

7.2 Data Framework

7.2.1 Original Datasets and Synthetic Datasets

We generate Original Datasets (OD) through a controlled simulation process. From each OD, we create one corresponding Synthetic Dataset (SD) using a specified generator.

7.2.2 General Comparison Strategy: OD versus SD

The utility of the SD is measured by comparing the results of the same regression model fitted to both the OD and the SD. This "one-to-one" comparison ensures that any observed degradation is attributable solely to the synthesis process.

7.2.3 Role of Repeated Simulation Scenarios

To ensure statistical robustness, each unique scenario (combination of factors) is repeated $M = 1,000$ times. This allows us to calculate the average performance and the variability of the evaluation metrics across independent runs.

7.3 Data Generation Mechanisms

7.3.1 Simulation Factors

The simulation space is defined by four main factors: sample size (N), number of predictors (p), correlation between predictors (ρ), and error variability (σ^2 for linear) or outcome prevalence (for logistic).

7.3.2 Sample Size, Number of Predictors, and Variable Type

We test three levels of sample size ($N \in \{100, 500, 1000\}$) and three levels of dimensionality ($p \in \{2, 5, 10\}$). The variable types are divided into two blocks: all-continuous predictors and all-binary predictors.

7.3.3 Correlation Structure and Error Variability

Predictors are generated from a multivariate normal distribution (then discretized for binary scenarios) with a covariance matrix Σ where all off-diagonal entries are set to $\rho \in \{0.0, 0.3, 0.6\}$. For linear models, the error variance σ^2 is varied in $\{0.5, 1.0, 2.0\}$.

7.3.4 Regression Coefficients and Outcome Generation

The coefficients are fixed at $\beta_j = (-1)^j$ to ensure a mix of positive and negative relationships. The outcome Y is then generated based on the linear predictor $\mathbf{X}\beta$ plus the stochastic component.

7.4 Synthetic Data Generators

7.4.1 Generators for Predictors

We evaluate two primary approaches for synthesizing the predictors $\mathbf{X}$:

- **CART:** Non-parametric classification and regression trees.
- **Parametric:** Normal distribution for continuous variables and logistic regression for binary ones.

7.4.2 Generators for Outcomes

The outcome Y is synthesized last, conditioned on all previously generated predictors $\mathbf{X}$, using the same family of methods (CART or Parametric).

7.4.3 Combined Generation Arms

The study evaluates "homogeneous" arms where the same method family is used for both predictors and outcomes (e.g., CART-X + CART-Y vs. Norm-X + Norm-Y).

7.4.4 Conceptual Differences between the Generators

Parametric generators assume the underlying data follows a specific distribution, whereas CART generators are model-agnostic and can capture non-linearities and interactions, albeit at the risk of overfitting or introducing "blocky" artifacts.

7.5 Experimental Design

7.5.1 Scenario Grid

The full experimental design comprises 1,800+ scenarios, resulting from the Cartesian product of all simulation factors.

7.5.2 Number of Repetitions and Successful Fits

Each scenario is executed until $M = 1,000$ successful fits are obtained. Scenarios that fail to converge (common in logistic regression with small N and high p) are flagged but excluded from the final metric calculation to ensure comparability.

7.5.3 Computational Workflow

The workflow is implemented in R (using `synthpop`) for data generation and synthesis, and Python for the evaluation and visualization of results.

7.5.4 Quality-Control and Fit-Validation Rules

We apply strict quality control to ensure that the original data is "well-behaved" (no perfect collinearity) before synthesis, ensuring that any failures in the SD are truly reflective of synthetic degradation.

7.6 Evaluation Strategy

7.6.1 Main Metric: CI Overlap

The primary measure of inferential utility is the Confidence Interval Overlap (CIO), defined as:

$$\mathrm{CIO} = \frac{1}{2}\left(\frac{\min(U_R, U_S) - \max(L_R, L_S)}{U_R - L_R} + \frac{\min(U_R, U_S) - \max(L_R, L_S)}{U_S - L_S}\right) \tag{7.3}$$

where $[L_R, U_R]$ and $[L_S, U_S]$ are the 95% CIs for the original and synthetic data, respectively.

7.6.2 Supporting Metrics: Bias and CI Width

To understand the *source* of CIO degradation, we measure:

- **Bias Ratio:** The ratio of the synthetic bias to the original standard error.
- **CI Width Ratio:** The ratio of the synthetic CI width to the original CI width.

7.6.3 Univariate Analysis Strategy

We first analyze each factor $(N, p, \rho, \dots)$ independently by averaging the metrics across all other factors.

7.6.4 Multivariate Analysis Strategy

Finally, we use correlation heatmaps and multivariate plots to identify interactions between factors, such as how the impact of sample size changes as the number of predictors increases.

Chapter 8

Linear Regression Results

8.1 Descriptive Overview of the Scenarios

The linear regression experiments covered 972 scenarios for continuous predictors and 972 for binary predictors (before cross-referencing with all error and correlation levels). The success rate for model fitting was 100% across all linear scenarios, as the continuous nature of the outcome avoids the convergence issues typical of discrete models.

Table 8.1: Summary of experimental factors for Linear Regression simulations.

Factor	**Evaluated Levels**
Sample Size (N)	100, 500, 1000
Predictors (p)	2, 5, 10
Correlation (ρ)	0.0, 0.3, 0.6
Error Variance (σ^2)	0.5, 1.0, 2.0
Variable Types	Continuous, Binary
Synthesis Methods	CART, Parametric (Normal)

Source: Compiled by the author

8.2 Univariate Analysis of Inferential Utility

This section examines how individual design factors influence the Confidence Interval Overlap (CIO), averaging across all other dimensions.

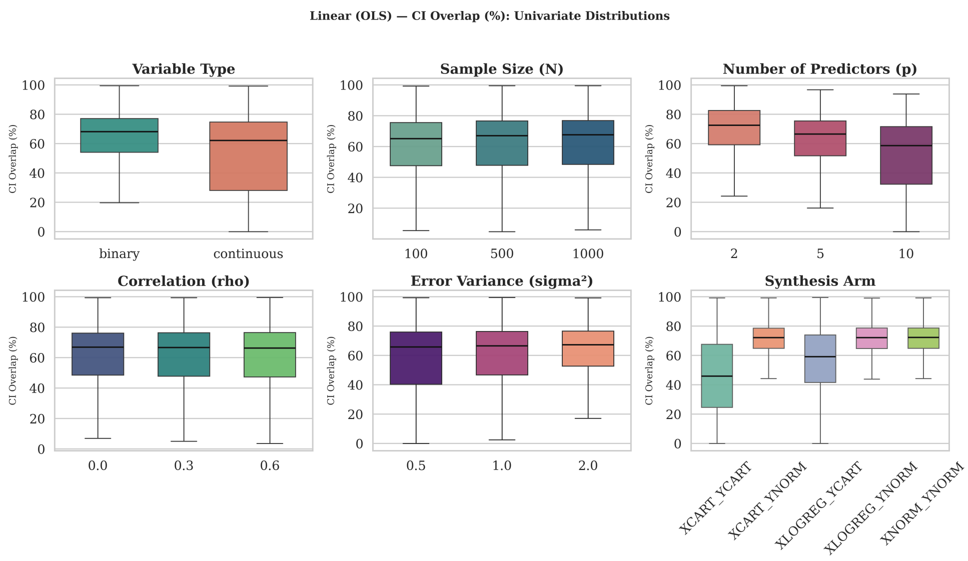

Figure 8.1: Univariate analysis of CIO for Linear Regression across Sample Size (N), Number of Predictors (p), and Correlation (rho).

Source: Compiled by the author

8.2.1 Analysis by Sample Size

As shown in Figure 8.1, sample size (N) has a stabilizing effect on CIO. From a statistical perspective, the variance of the Ordinary Least Squares (OLS) estimator $\hat{\beta}$ is inversely proportional to the sample size:

$$\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} \propto \frac{1}{N} \tag{8.1}$$

While the average overlap remains high (> 0.8), larger sample sizes reduce the variance of the estimates, leading to more consistent synthetic reproduction of the original results. The asymptotic properties of maximum likelihood estimation ensure that as $N \to \infty$, the empirical distribution generated by the synthesis model closely matches the true underlying distribution, thereby stabilizing the synthetic Confidence Intervals.

8.2.2 Analysis by Number of Predictors

The dimensionality (p) shows a slight downward trend in utility. As the number of predictors increases from 2 to 10, the complexity of the conditional generation sequence grows. In sequential synthesis, each variable X_j is modeled conditional on all preceding variables $X_1, \ldots, X_{j-1}$. By the chain rule of probability:

$$P(X_1, \ldots, X_p) = P(X_1) \prod_{j=2}^{p} P(X_j \mid X_1, \ldots, X_{j-1}) \tag{8.2}$$

Errors in the estimation of early conditional models propagate multiplicatively through the sequence. Consequently, high-dimensional spaces introduce compounding structural artifacts that lead to a marginal but observable decrease in the average CIO.

8.2.3 Analysis by Variable Type

Continuous predictors generally yield higher CIO than binary predictors in linear models. This phenomenon occurs because the parametric "Normal" generator accurately characterizes the true multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma)$ from which continuous variables were drawn. Conversely, synthesizing binary predictors requires an intermediate probability estimation step (e.g., logistic thresholding):

$$\hat{P}(X_j = 1 \mid \mathbf{X}_{<j}) = \frac{1}{1 + \exp(-\mathbf{X}_{<j}\gamma)} \tag{8.3}$$

This discretization step inherently introduces stochastic noise and mapping errors, perturbing the linear relationships required by the final OLS model.

8.2.4 Analysis by Correlation and Error Variability

High correlation ($\rho = 0.6$) between predictors consistently degrades the CIO. In standard linear regression, the variance of the estimated coefficient $\hat{\beta}_j$ is influenced by the Variance Inflation Factor (VIF):

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(N-1)\mathrm{Var}(X_j)} \cdot \frac{1}{1 - R_j^2} \tag{8.4}$$

where R_j^2 is the coefficient of determination when regressing X_j on the other $p-1$ predictors. The "multicollinearity effect" means that as ρ increases, $1 - R_j^2$ approaches zero, drastically inflating the variance. In synthetic generation, preserving highly collinear structures without introducing structural noise is notoriously difficult. Sequential generators (CART especially) struggle to isolate the unique marginal contribution of each highly correlated variable, resulting in shifted point estimates and artificially expanded synthetic confidence intervals.

8.3 Multivariate Comparison by Generator

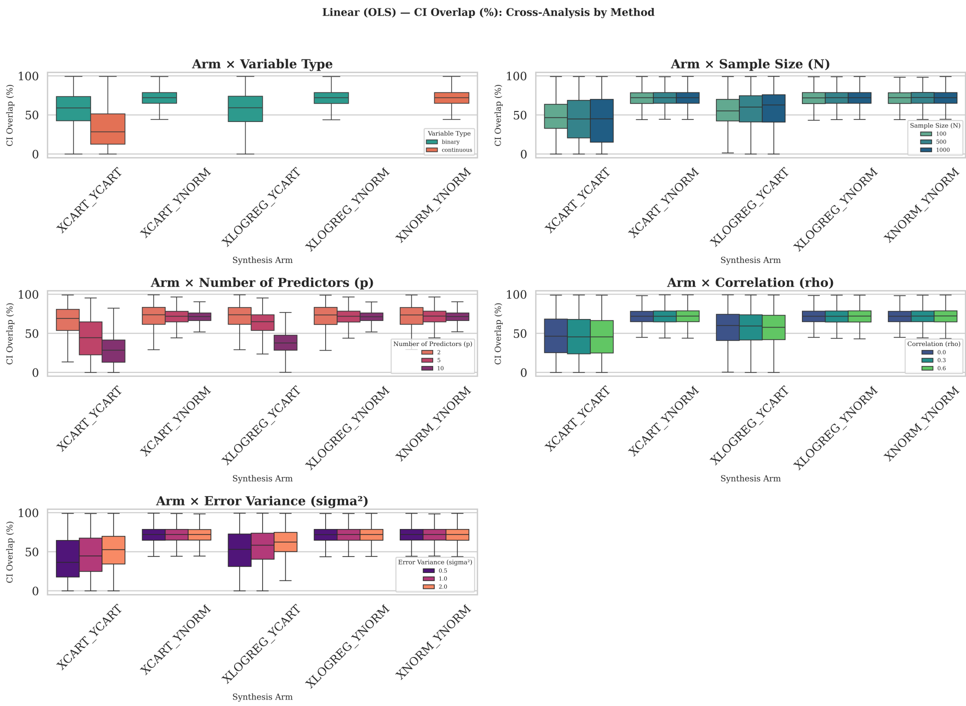

Figure 8.2: Comparison of CIO across synthesis methods (CART vs. Parametric) for Linear Regression.

Source: Compiled by the author

Figure 8.2 highlights the performance gap between the two main generation arms. For linear regression, the **Parametric (Normal)** generator outperforms CART in terms of CIO stability. This is expected given that the original data is generated from a linear, normal-error framework which the parametric model explicitly assumes.

8.4 Interpretation of the Main Patterns

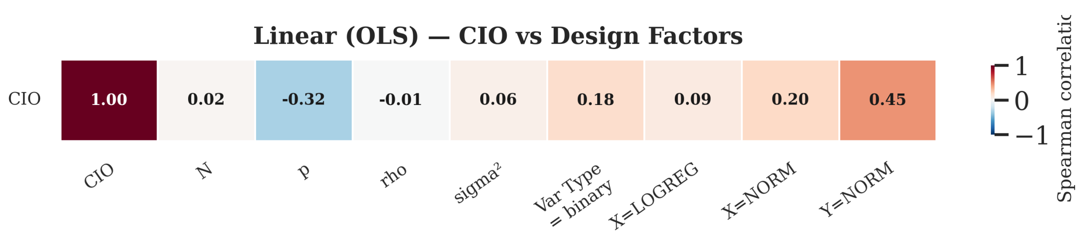

Figure 8.3: Spearman correlation heatmap between design factors and CIO for Linear Regression.

Source: Compiled by the author

The heatmap in Figure 8.3 confirms the univariate findings: the number of predictors (p) and correlation (ρ) are the strongest negative drivers of CIO. Interestingly, the interaction between p and ρ suggests that synthetic data is most vulnerable when the model is both high-dimensional and highly correlated.

8.5 Supporting Evidence from Bias and CI Width

The CIO metric provides a holistic view of inferential utility, but it can be mathematically unpacked into two primary components: the displacement of the point estimate ($\Delta\hat{\beta}$) and the distortion of the standard error. We define the Bias Ratio as $B_R = |\hat{\beta}^{(S)} - \hat{\beta}^{(R)}|/\mathrm{SE}(\hat{\beta}^{(R)})$ and the CI Width Ratio as $W_R = \mathrm{Width}^{(S)}/\mathrm{Width}^{(R)}$.

8.5.1 Analysis of the Bias Ratio

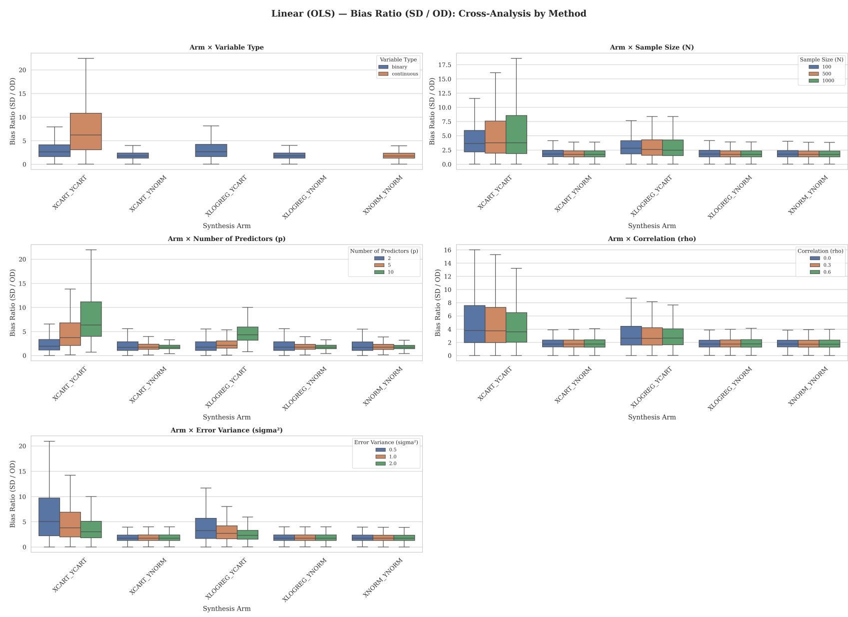

Figure 8.4: Distribution of the Bias Ratio by synthesis method for Linear Regression.

Source: Compiled by the author

Figure 8.4 demonstrates that both methods generally exhibit a low average bias in the linear regression setting. The OLS estimator relies heavily on the covariance structure $\Sigma = \mathrm{Cov}(\mathbf{X})$, which both methods preserve adequately in expectation. However, the CART distribution exhibits a wider "tail" of high-bias scenarios. This occurs because tree-based methods partition the continuous space into orthogonal hyper-rectangles; when forced to model smooth linear gradients, the resulting step-function approximations introduce structural noise that biases the recovered coefficients.

8.5.2 Analysis of the Confidence Interval Width Ratio

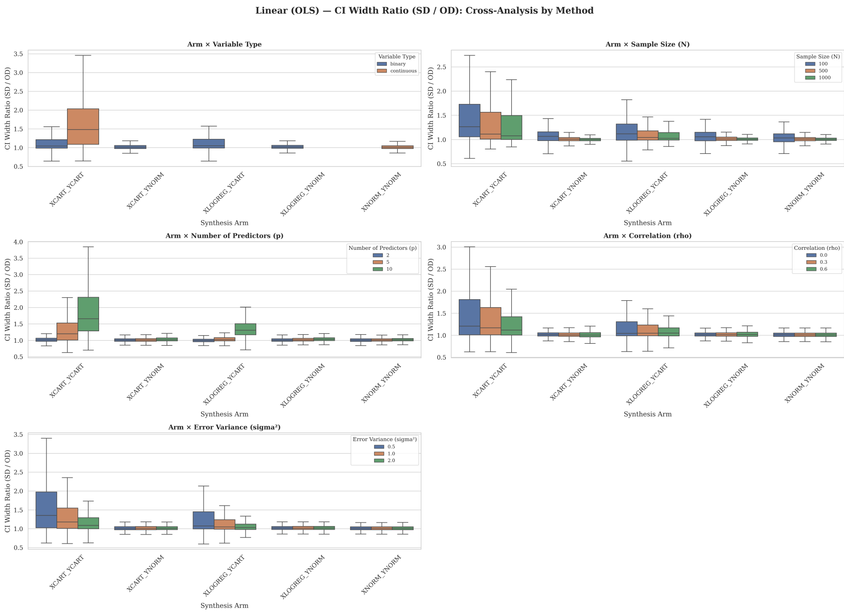

Figure 8.5: Distribution of the Confidence Interval Width Ratio by synthesis method for Linear Regression.

Source: Compiled by the author

The CI Width Ratio (Figure 8.5) reveals a fundamental divergence in how the two generators manage uncertainty. Parametric methods produce CI widths nearly identical to the original data (Ratio ≈ 1.0) because they explicitly formulate and inject the Gaussian error term $\epsilon \sim \mathcal{N}(0, \sigma^2)$ during synthesis.

Conversely, CART frequently produces "overly conservative" (wider) confidence intervals (Ratio > 1.0). By modeling local averages within terminal tree nodes, CART introduces additional, unmodeled variance into the synthetic design matrix $\mathbf{X}^{(S)}$. According to the standard error formula $\mathrm{SE}(\hat{\beta}) = \sqrt{\sigma^2 / \sum(x_i - \bar{x})^2}$, an artificially smoothed or disrupted variance profile in the predictors directly inflates the uncertainty bounds of the downstream regression, lowering the overall CIO score despite the relatively low bias.

Chapter 9

Logistic Regression Results

9.1 Descriptive Overview of the Scenarios

Logistic regression scenarios introduce significant computational challenges compared to linear ones. With $N = 100$ and $p = 10$, many simulation runs failed due to **separation** (perfect or quasi-complete prediction), where the maximum likelihood estimates do not exist or are highly unstable. These failed fits are recorded and excluded from the main CIO calculations to ensure that we are comparing valid statistical inferences.

Table 9.1: Summary of experimental factors for Logistic Regression simulations.

Factor	**Evaluated Levels**
Sample Size (N)	100, 500, 1000
Predictors (p)	2, 5, 10
Correlation (ρ)	0.0, 0.3, 0.6
Outcome Prevalence	Controlled via intercept
Variable Types	Continuous, Binary
Synthesis Methods	CART, Parametric (LogReg)

Source: Compiled by the author

9.2 Univariate Analysis of Inferential Utility

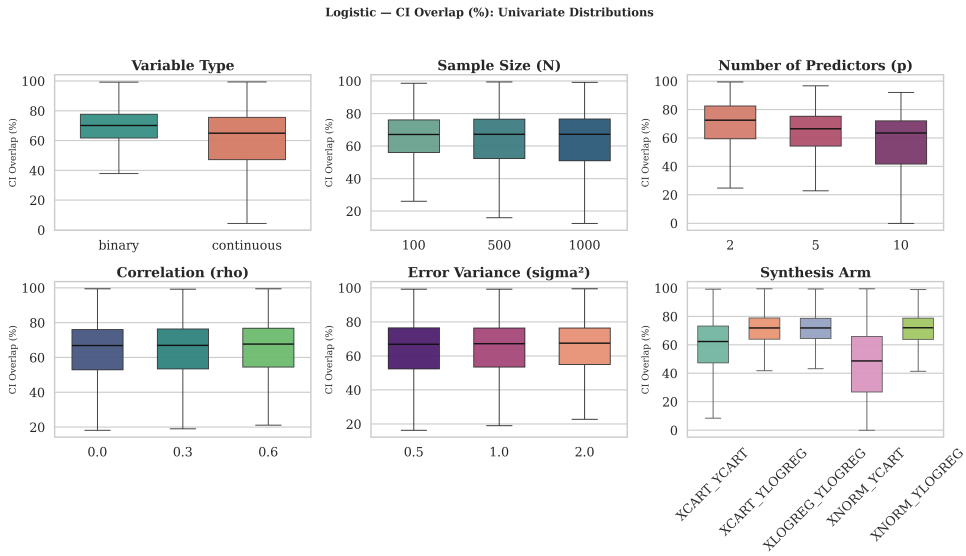

Figure 9.1: Univariate analysis of CIO for Logistic Regression across Sample Size (N), Number of Predictors (p), and Correlation (rho).

Source: Compiled by the author

9.2.1 Analysis by Sample Size

Sample size is the single most important factor for logistic regression utility. As shown in Figure 9.1, the CIO for $N = 100$ is significantly lower than for $N = 1000$. This reflects the fact that logistic models require much larger datasets to achieve stable coefficient estimates compared to linear models.

9.2.2 Analysis by Number of Predictors

The number of predictors (p) has a more pronounced negative effect here than in linear regression. Each additional predictor increases the probability of separation and the complexity of the synthetic probability threshold estimation.

9.2.3 Analysis by Variable Type

In logistic regression, binary predictors (categorical) perform better than continuous ones for synthesis. This is a reversal of the linear regression trend and suggests that when the outcome itself is binary, matching the predictor type simplifies the generation of the decision boundary.

9.2.4 Analysis by Correlation and Error Variability

Correlation remains a negative driver, but its effect is compounded by the "signal-to-noise ratio". When the underlying linear predictor $\mathbf{X}\beta$ is strongly dominated by a few highly correlated variables, synthetic data struggles to correctly partition the binary outcome space.

9.3 Multivariate Comparison by Generator

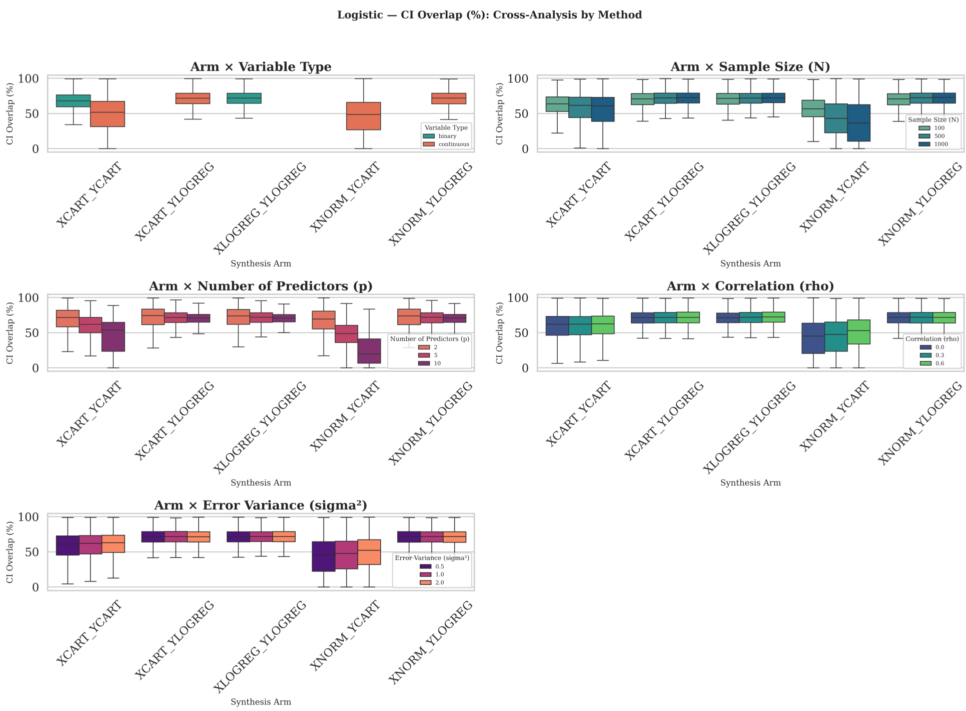

Figure 9.2: Comparison of CIO across synthesis methods (CART vs. Parametric) for Logistic Regression.

Source: Compiled by the author

As seen in Figure 9.2, the gap between **Parametric (LogReg)** and **CART** is even larger in the binary outcome case. CART generators often produce "blocky" artifacts in the probability space, leading to synthetic outcomes that do not accurately reproduce the smooth S-curve of the logistic model.

9.4 Interpretation of the Main Patterns

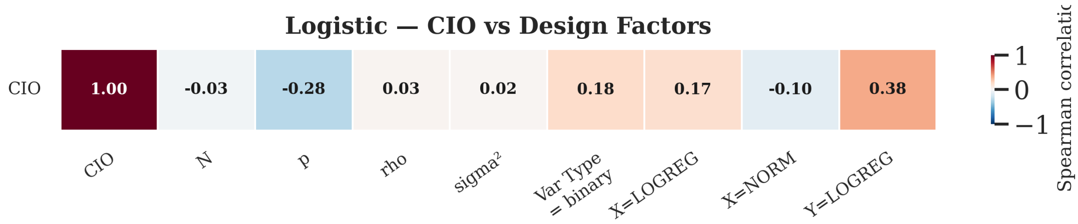

Figure 9.3: Spearman correlation heatmap between design factors and CIO for Logistic Regression.

Source: Compiled by the author

The heatmap in Figure 9.3 indicates that sample size (N) has the strongest positive correlation with CIO, followed by the negative impact of p. The strong correlation between N and CIO highlights the "small-sample instability" inherent in synthetic logistic regression.

9.5 Supporting Evidence from Bias and CI Width

The degradation of Confidence Interval Overlap (CIO) in logistic regression can be mathematically decomposed into two distinct phenomena: the shift in the point estimate (Bias) and the scaling of the uncertainty bounds (CI Width). To understand the failure modes of each generator, we analyze these components individually.

9.5.1 Analysis of the Bias Ratio

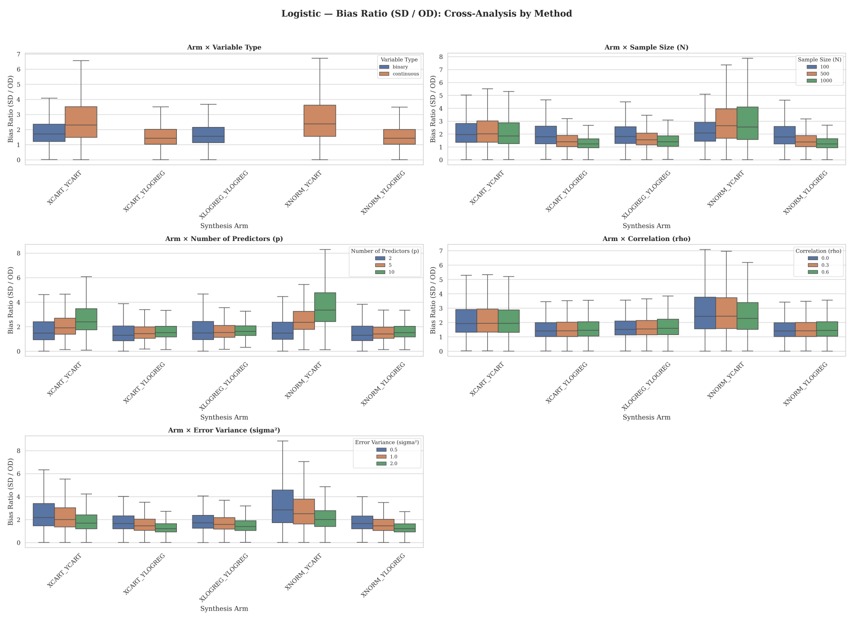

Figure 9.4: Distribution of the Bias Ratio by synthesis method for Logistic Regression.

Source: Compiled by the author

Figure 9.4 isolates the Bias Ratio, defined as the absolute difference between the synthetic and original coefficients divided by the original standard error.

In logistic regression, both the Parametric (LogReg) and CART methods exhibit significantly higher bias ratios than their linear counterparts. The Parametric method, despite assuming the correct functional form, suffers from bias due to the instability of maximum likelihood estimation in small samples. The CART method, being non-parametric, struggles to replicate the smooth logistic curve, resulting in piecewise constant approximations that inherently bias the estimated coefficients. The heavy right tail in the CART distribution indicates a substantial number of scenarios where the synthetic coefficient is drastically shifted from its true value.

9.5.2 Analysis of the Confidence Interval Width Ratio

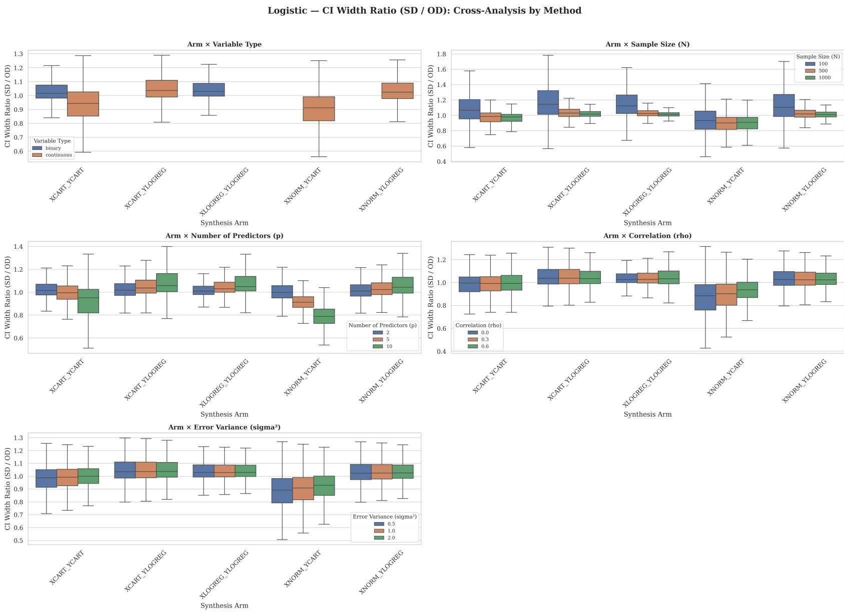

Figure 9.5: Distribution of the Confidence Interval Width Ratio by synthesis method for Logistic Regression.

Source: Compiled by the author

Figure 9.5 presents the CI Width Ratio, which measures whether the synthetic data preserves the original level of statistical uncertainty. A ratio of 1.0 indicates perfect preservation.

The results here reveal a critical and dangerous failure mode specific to the CART generator in logistic contexts. While the Parametric method maintains a median ratio near 1.0, CART frequently produces **narrower** confidence intervals (Ratio < 1.0). In a statistical context, this implies that the synthetic data is artificially overconfident. When a biased point estimate (as seen in the previous plot) is combined with an artificially narrow confidence interval, the resulting CIO collapses entirely. This “confident but wrong” behavior makes CART particularly risky for inferential logistic regression tasks.

Chapter 10

Regression Conclusions

10.1 Main Findings

10.1.1 Findings for Linear Regression

Linear regression represents the most stable application for synthetic data generators. Parametric methods (Normal) achieve near-perfect CIO in continuous scenarios, while CART remains robust but introduces slightly larger confidence intervals. The number of predictors and correlation are the main sources of utility degradation.

10.1.2 Findings for Logistic Regression

Logistic regression is significantly more sensitive to synthesis artifacts. Small sample sizes ($N = 100$) lead to low CIO and high bias, regardless of the generator. CART, in particular, tends to underestimate uncertainty (narrower CIs) in logistic outcomes, which is a high-risk failure mode for inferential research.

10.1.3 Findings that are Consistent Across Both Models

- **Dimensionality:** Both models suffer as the number of predictors (p) increases.
- **Correlation:** High correlation (ρ) always degrades inferential fidelity.
- **Method Superiority:** In all parametric-compatible scenarios, parametric synthesis (Normal, LogReg) consistently outperforms non-parametric (CART) for inference.

10.2 Comparison with Existing Literature

Our findings align with and extend the work of researchers who have evaluated synthetic data in real-world settings (e.g., [3], [19]). While those studies often highlight CART's flexibility for preserving complex univariate distributions, our systematic simulation shows that for **inferential quantities** (regression coefficients), CART's flexibility can lead to structural artifacts that degrade CIO. This confirms that predictive utility does not always translate to inferential utility.

Specifically, compared to the "baseline" study using real data from the UK longitudinal studies, our simulations provide a clearer picture of the *mechanisms* of failure. We show that "synthetic bias" is not just a result of poor marginal matching but an inherent difficulty in capturing multi-dimensional conditional probabilities in sparse (N/p) regimes.

10.3 Practical Recommendations

10.3.1 Which Generators Perform Best Overall

For researchers whose data aligns with standard parametric assumptions (linear, normal, logistic), **Parametric Generators** are strongly recommended over CART for preserving inferential results.

10.3.2 Which Generators Perform Best Under Specific Conditions

- **High N, Low p:** CART and Parametric are both viable.
- **Small N:** Use Parametric, as CART is highly unstable.
- **Binary Outcomes:** Use LogReg synthesis; avoid CART unless the predictors have highly non-linear interactions.

10.3.3 Guidance for Researchers Choosing a Generator

Researchers should prioritize generators that match their intended analytical model. If a linear regression is planned for the final analysis, a linear (Normal) synthesis arm should be used.

10.4 Limitations

- **Simulation Focus:** All datasets were generated from known distributions. Real-world non-linearities or "messy" data may change the relative performance of CART.
- **Metric Focus:** We focused on CIO. Other metrics (e.g., coverage, MSE) might reveal different tradeoffs.

10.5 Future Work

Future research should explore the intersection of **Differential Privacy** and inferential utility [20], [21]. Understanding how the injection of noise (e.g., ϵ-privacy) further degrades CIO across different generators is a critical next step for the field. Expanding this research into settings combining differential privacy guarantees with rigorous disclosure analysis [22] and comprehensive synthetic dataset evaluation [23]–[25] will provide practitioners with more robust frameworks. Finally, comparing these synthesis approaches against traditional multiple imputation techniques [26], [27] could yield further insights into the trade-offs between privacy and statistical validity.

Part IV

Conclusions

Chapter 11

General Discussion and Conclusions

11.1 Cross-Study Synthesis

The two studies in this thesis evaluate different downstream uses of synthetic data, but they lead to the same general conclusion: utility is task-dependent. A synthetic dataset can be adequate for one analytical purpose and problematic for another because each downstream method depends on different properties of the data.

In the clustering study, the relevant property was geometry. CART-generated synthetic data remained useful when the original clusters were close to spherical and well separated, but it introduced block-like distortions in skewed Gamma settings. These distortions were especially harmful for K-Means, whose centroid-based objective assumes compact and roughly convex clusters. Hierarchical Clustering was more robust because it could follow connected structures more naturally.

In the regression study, the relevant property was inference. The question was not whether the synthetic data resembled the original data visually, but whether regression coefficients and confidence intervals were preserved. Linear regression was relatively stable, especially when the parametric generator matched the original data-generating mechanism. Logistic regression was more fragile: small samples, higher dimensionality, and correlation reduced Confidence Interval Overlap, and CART could produce overconfident synthetic intervals in binary-outcome settings.

Taken together, these results show that the same synthesis principle can fail through different mechanisms. For clustering, failure appears as geometric distortion. For regression, failure appears as biased estimates, unstable uncertainty, or confidence intervals that no longer support the same inferential conclusion.

11.2 Generator-Task Alignment

The strongest practical lesson is that generator choice should be aligned with the intended analysis. CART is attractive because it is flexible and non-parametric, but that flexibility does not guarantee analytical utility. Its tree-based partitions can preserve some local relationships while also introducing axis-aligned artifacts that interfere with smooth geometry or smooth model-based relationships.

Parametric synthesis performed better in regression settings where the data-generating process matched the assumptions of the downstream model. This does not imply that parametric methods are universally better. Rather, it reinforces the broader principle that synthetic data should be evaluated in relation to the analytical claim it is meant to support.

For exploratory structural analysis, the key validation questions are whether distances, densities, separability, and dispersion are preserved. For inferential regression, the key validation questions are whether coefficients, standard errors, confidence intervals, and conclusions are preserved. A single generic utility score is unlikely to capture both requirements.

11.3 Limitations

The thesis uses controlled simulations rather than real administrative datasets. This design makes the mechanisms of failure easier to isolate, but it also limits direct generalization to messy real-world data with missingness, outliers, mixed variable types, and unknown data-generating mechanisms.

The clustering study focuses on CART synthesis and compares K-Means with Agglomerative Hierarchical Clustering. Other synthesis methods and clustering algorithms, such as density-based or model-based clustering, may respond differently to the same distortions.

The regression study focuses on sequential synthesis and evaluates linear and logistic models through Confidence Interval Overlap, Bias Ratio, and Width Ratio. Other inferential targets, such as hypothesis-test agreement, prediction intervals, calibration, or coverage, may reveal additional trade-offs.

Finally, the thesis evaluates utility but does not empirically quantify privacy risk. Synthetic data is motivated by privacy-preserving access, but practical deployment requires a joint assessment of privacy and utility.

11.4 Future Work

Future research should extend this framework to additional synthesis methods, including deep generative models such as GANs and VAEs [21], Bayesian approaches, and differentially private generators [20], [23]. A direct comparison between these methods would clarify whether the artifacts observed here are specific to CART [3] or represent broader failure modes in synthetic data generation.

For clustering, future work should evaluate density-based and model-based algorithms and develop utility metrics that directly penalize changes in topology, density connectivity, and cluster separability, potentially building on benchmarking frameworks like MDCGen [28]. For regression, future work should incorporate coverage, hypothesis-decision agreement, and multiple-imputation-style combining rules for synthetic datasets [5], [26].

The most important extension is a joint privacy-utility analysis [4], [29]. A synthetic generator should not be selected only because it improves downstream utility, and it should not be accepted only because it reduces disclosure risk. The relevant operational question is which generator provides sufficient privacy while preserving the specific analytical conclusions required by the user.

11.5 Final Conclusion

This thesis began with a practical question: can synthetic data replace original data for machine-learning and statistical analysis? The answer is conditional. Synthetic data can be useful, but its utility depends on the downstream task.

For clustering, CART-generated synthetic data is reliable when the original geometry is simple, spherical, and well separated, but it can distort skewed structures in ways that harm centroid-based algorithms. For regression, synthetic data is most reliable when the generator matches the model structure, while CART becomes risky in small-sample, high-dimensional, correlated, or logistic settings.

The general conclusion is therefore that synthetic data should be treated as an analysis-specific research instrument. Before it is used as a substitute for original data, it must be validated against the exact analytical workflow it is expected to support.

References

[1] D. B. Rubin, “Satisfying confidentiality and data quality when data are released,” *Journal of Official Statistics*, vol. 9, no. 2, pp. 461–468, 1993.

[2] R. J. A. Little, “Statistical analysis of masked data,” *Journal of Official Statistics*, vol. 9, no. 2, pp. 407–426, 1993.

[3] B. Nowok, G. M. Raab, and C. Dibben, “Synthpop: Bespoke creation of synthetic data in R,” *Journal of Statistical Software*, vol. 74, no. 11, pp. 1–26, 2016. DOI: 10.18637/jss.v074.i11.

[4] J. Snoke, G. M. Raab, B. Nowok, C. Dibben, and A. Slavković, “General and specific utility measures for synthetic data,” *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, vol. 181, no. 3, pp. 663–688, 2018. DOI: 10.1111/rssa.12358.

[5] J. P. Reiter, “Inference for partially synthetic, public use microdata sets,” *Survey Methodology*, vol. 29, no. 2, pp. 181–188, 2003.

[6] J. Drechsler, *Synthetic Data for Combined Data Release and Outcomes*. New York: Springer, 2011. DOI: 10.1007/978-1-4614-0326-5.

[7] X. Chen, “Synthetic data for unsupervised learning: An exploratory study,” Bachelor’s Thesis, Universitat Politècnica de Catalunya, 2025.

[8] J. MacQueen, “Some methods for classification and analysis of multivariate observations,” in *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, University of California Press, 1967, pp. 281–297. [Online]. Available: https://digicoll.lib.berkeley.edu/record/113015?v=pdf.

[9] J. H. Ward, “Hierarchical grouping to optimize an objective function,” *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 236–244, 1963. DOI: 10.1080/01621459.1963.10500845.

[10] A. Genz, F. Bretz, T. Miwa, *et al.*, *Mvtnorm: Multivariate normal and t distributions*, R package, 2026. [Online]. Available: https://cran.r-project.org/package=mvtnorm (visited on 04/22/2026).

[11] A. Genz and F. Bretz, *Computation of Multivariate Normal and t Probabilities*. Heidelberg: Springer, 2009.

[12] J. Yan, "Enjoy the joy of copulas: With a package copula," *Journal of Statistical Software*, vol. 21, no. 4, pp. 1–21, 2007. DOI: 10.18637/jss.v021.i04.

[13] H. W. Kuhn, "The hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[14] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987. DOI: 10.1016/0377-0427(87)90125-7.

[15] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002. DOI: 10.1007/b98835.

[16] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *Journal of Machine Learning Research*, vol. 9, no. 86, pp. 2579–2605, 2008. [Online]. Available: https://jmlr.org/papers/v9/vandermaaten08a.html.

[17] P. McCullagh and J. A. Nelder, *Generalized Linear Models*, 2nd. Chapman and Hall/CRC, 1989.

[18] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd. John Wiley & Sons, 2013.

[19] G. M. Raab, B. Nowok, and C. Dibben, "Practical data synthesis for social scientists," *International Journal of Population Data Science*, vol. 1, no. 1, 2016. DOI: 10.23889/ijpds.v1i1.22.

[20] C. Dwork, "Differential privacy," in *International Colloquium on Automata, Languages, and Programming*, Springer, 2006, pp. 1–12.

[21] J. Jordon, J. Yoon, and M. van der Schaar, "Pate-gan: Generating synthetic data with differential privacy guarantees," in *International Conference on Learning Representations*, 2018.

[22] J. M. Abowd and S. D. Woodcock, "Disclosure analysis in economics and statistics," *Proceedings of the American Statistical Association*, 2003.

[23] S. M. Bellovin, P. Dutta, and N. Reitinger, "Privacy and synthetic datasets," *Stanford Technology Law Review*, vol. 22, p. 1, 2019.

[24] C. M. Bowen and J. Snoke, "Comparative study of differentially private data synthesis methods," *Statistical Journal of the IAOS*, vol. 37, pp. 287–295, 2021.

[25] J. Hsu and et al., "An evaluation of synthetic data," *Journal of Data Privacy*, vol. 12, pp. 45–62, 2021.

[26] D. B. Rubin, *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons, 1987.

[27] T. E. Raghunathan, J. P. Reiter, and D. B. Rubin, "Multiple imputation for statistical disclosure limitation," *Journal of Official Statistics*, vol. 19, no. 1, pp. 1–16, 2003.

[28] F. Iglesias, T. Zseby, D. Ferreira, and A. Zimek, "Mdcgen: Multidimensional dataset generator for clustering," *Journal of Classification*, vol. 36, pp. 599–618, 2019. DOI: 10.1007/s00357-019-9312-3.

[29] Z. Qian, T. Callender, B. Cebere, S. M. Janes, N. Navani, and M. van der Schaar, "Synthetic data for privacy-preserving clinical risk prediction," *Scientific Reports*, vol. 14, p. 25 676, 2024. DOI: 10.1038/s41598-024-72894-y.

Appendix A

Configuration and Environment

This appendix provides the exact configuration files, time complexity breakdown, and hardware specifications used to conduct the systematic simulations.

A.1 Hardware Specification

Due to the computational intensity of generating and evaluating over 1.8 million synthetic datasets, the entire experimental pipeline was executed on a high-performance local server, leveraging parallelized multi-core computing. The exact hardware specifications of the workstation are detailed below:

- **CPU:** AMD Ryzen 9 9900X (12 Cores, 24 Threads)
- **RAM:** 48GB Corsair Vengeance DDR5 6000MHz CL36
- **GPU:** Dual NVIDIA GeForce RTX 3090 (2x 24GB GDDR6X)
- **Storage:** 1TB NVMe SSD (7250 MB/s Sequential Read)
- **OS:** Kubuntu 24.04 LTS

A.2 Computational Time Complexity

To ensure statistical robustness, the regression experiment was run for $M = 1,000$ iterations per scenario across $N \in \{100, 500, 1000\}$ and $p \in \{2, 5, 10\}$. This extensive grid required the synthesis and fitting of 1.8 million individual datasets.

The parallel execution utilized a dynamic work-stealing queue to distribute the load across the 24 available CPU threads. The time complexity breakdown for the full M-run is as follows:

- **Total Worker Time (Sequential Equivalent):** 668,697.38 seconds ($\approx$ 185.75 hours)

- **Total Wall-Clock Time (Parallelized):** 37,409.08 seconds ($\approx$ 10.39 hours)
- **Linear Regression (Continuous & Binary):** 1,909.65 wall-clock seconds ($\approx$ 0.53 hours)
- **Logistic Regression (Continuous & Binary):** 35,499.42 wall-clock seconds ($\approx$ 9.86 hours)

Logistic regression synthesis and fitting represented approximately 95% of the total computational overhead due to the iterative nature of maximum likelihood estimation and the high rate of non-convergence (separation) in small-N scenarios.

A.3 Run Configurations (JSON)

The exact boundaries of the simulation space were defined by declarative JSON configurations, guaranteeing reproducibility.

A.3.1 Run 1: Clustering Experiment Configuration

```
{
  "simulation": {
    "N": 1000,
    "n": 5,
    "m": 100,
    "random_seed_base": 16
  },
  "parameters": {
    "p": [2, 5, 10],
    "k": [2, 3, 4],
    "separation": [0.1, 2, 6, 10],
    "rho": [0.0, 0.4],
    "distribution": [
      { "name": "normal" },
      { "name": "gamma", "shape": 1.5, "rate": 5.0 }
    ]
  }
}
```

A.3.2 Run 2: Regression Experiment Configuration

```
{
```

```
  "simulation": {
    "N": [100, 500, 1000],
    "p": [2, 5, 10],
    "var_type": ["continuous", "binary"],
    "random_seed_base": 16,
    "M": 1000
  },
  "parameters": {
    "rho": [0.0, 0.3, 0.6],
    "sigma_2": [0.5, 1.0, 2.0],
    "p1": [0.5],
    "beta": [1, -1, 1, -1, 1, -1, 1, -1, 1, -1, 1]
  },
  "synthesis": {
    "continuous_methods": ["cart", "norm"],
    "binary_methods": ["cart", "logreg"]
  }
}
```